MAHROS: A Multi-Agent Hospital Resource Optimization System for Verifiable, Privacy-Preserving Inter-Hospital Transfer Negotiation

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***Abstract*—When an admitted patient deteriorates beyond what the admitting hospital can provide (an ICU bed, a catheterisation laboratory, a neurosurgeon), the patient must be transferred, and today the receiving hospital is found by a coordinator telephoning hospitals one at a time. The search is serial and slow, it consumes scarce staff time, and a refusal such as "we are full" cannot be checked by anyone. Centralised bed registries remove the serial search but require every hospital to disclose its occupancy and still believe whatever they are told. This paper presents MAHROS, a Multi-Agent Hospital Resource Optimization System in which one software agent per hospital, with one resource agent per bed pool, answers a Contract-Net call for proposals from its own private state, and every refusal becomes a claim that can be verified. Capability refusals are checked against the public CMS Provider of Services register. Capacity refusals are checked against the refusing hospital's own recent acceptances, recorded on a hash-chained ledger, so no hospital ever discloses a bed count. Disputes are resolved with Dung's grounded semantics, challenged hospitals may be required to sign their refusal with Ed25519, and random spot checks counter coordinated refusal. The system is implemented in Python with a FastAPI console and a PostgreSQL backend whose reference data are trigger-locked and whose every change is written to a hash-chained audit log. It is evaluated on 27 real Houston and 30 real Phoenix hospitals built from U.S. HHS capacity reports, CMS capability data, OpenStreetMap road times and 37,439 real California transfers. The calibrated simulator reproduces reported occupancy within 3.8 and 3.0 percentage points from a cold start. MAHROS places 13.0 points more patients within their clinical window than a perfect-information telephone search ($p < 10^{-4}$), is statistically equivalent within ± 2 points to optimisers that read every hospital's true occupancy, recovers 62–77% of the placements lost to false refusals with zero false accusations, outperforms a shared registry by up to 17.9 points under misreporting, and decides a transfer in a median of 3.6 ms.**

***Index Terms*—Multi-agent systems, contract net protocol, abstract argumentation, inter-hospital transfer, healthcare resource allocation, strategic misreporting, tamper-evident ledger, privacy-preserving coordination, discrete-event simulation.**

I. Introduction

A. Background

A patient admitted to a community hospital can deteriorate in ways that hospital is not equipped to manage. A suspected ST-elevation myocardial infarction needs a catheterisation laboratory, an intracranial bleed needs neurosurgery, and respiratory failure needs a staffed ICU bed and a ventilator. When the admitting hospital has none of these free, the patient must be moved to one that does. Such *escalation transfers* are routine: in a national Medicare cohort about 4.5% of critical-care stays involved an inter-hospital transfer, and transfers flowed along a structured hub-and-spoke network of community hospitals and tertiary centres [7]. Time matters. Among deteriorating ward patients, each hour of delay before ICU transfer was associated with about a 3% increase in the adjusted odds of death [8]. Delay is mostly organisational rather than geographical. Across 49 UK hospitals, transfer delays correlated only weakly with distance [9], and transfer processes such as night-time arrival and acceptance-to-arrival delay have been associated with worse outcomes [10].

The organisational process is, in most regions, a telephone search. A coordinator at the sending hospital rings the nearest plausible receiver, describes the patient, waits for an answer, and on a refusal moves to the next hospital. Regional transfer centres and shared bed boards, expanded rapidly during the COVID-19 pandemic [11], replace the telephone with a central dispatcher that reads every hospital's reported occupancy. A growing operations-research literature proposes optimisers that redistribute patients given full knowledge of each hospital's state [12], [13], [51].

B. Motivation

Both approaches share a weakness that neither can repair: they must take a hospital's word about its own capacity. Accepting a transfer costs the receiver a scarce bed for days and earns it little, so the cheapest response to an unwelcome request is "sorry, we are full". Evidence that capacity drives refusals is direct. In a study of 26,020 transfer requests, monthly acceptance at the receiving system was strongly and negatively correlated with its inpatient census ($r = -0.87$) and emergency-department boarding ($r = -0.73$) [14]. Whether a particular refusal was true cannot be established from outside, by phone or by registry. Self-reported capacity data are also demonstrably unreliable. The facility-level capacity data U.S. hospitals reported to the Department of Health and Human Services were the subject of a federal review that called for external validation [1], and Section VIII-D shows that 5.4%

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Source code, data and experiment scripts: https://github.com/ROHITH-V-S/MAHROS.

of the weekly reports from the 27 Houston hospitals studied here are impossible or look copied.

The motivation of MAHROS is therefore threefold. The search must become *parallel*, asking every capable hospital at once instead of one at a time. Occupancy must remain *private*, because hospitals will not hand live operational state to a competitor or to a central store. And a refusal must become *verifiable*: something a hospital can be held to, without the verifier ever seeing the hospital's beds, and without punishing a hospital whose refusal is true.

C. Scope of the Project

MAHROS covers the design, formal modelling, implementation and empirical evaluation of a decentralised transfer-brokering system for patients who are *already admitted*, within one metropolitan region. In scope are the following.

- A multi-agent architecture with hospital, resource, transfer-manager, verifier, data-integrity and regional-monitor agents.
- A Contract-Net negotiation protocol extended with a deliberation phase in which refusals are verified against public evidence and disputes are settled by grounded argumentation semantics.
- Signed (attested) refusals and random spot checks against coordinated refusal.
- A tamper-evident transfer ledger and a PostgreSQL back-end with trigger-locked reference data and a hash-chained audit log.
- A browser console that streams every protocol step.
- A discrete-event simulator calibrated to 57 real hospitals in Houston and Phoenix.
- A pre-registered statistical evaluation against five baselines.

Out of scope are ambulance dispatch, emergency-department triage, repatriation and billing, integration with commercial bed-management systems, and live clinical deployment. Hospital behaviour, including whether a refusal is truthful, is simulated because no public dataset records it, and every behavioural parameter is swept.

D. Contributions

1) **Verifiable refusals without disclosure.** A protocol that classifies each refusal as a public *capability* claim or a private *capacity* claim and verifies each against a different evidence base: the federal capability register, or the refuser's own ledger-recorded commitments. No message ever carries a bed count (Section V-F).
2) **Evidence-gated argumentation.** A four-rule attack rulebook over Dung argumentation frameworks in which an attack is admitted only when a public record supports it, so a truthful hospital always wins its dispute (Section V-G).
3) **Signed refusals and spot checks.** Ed25519-attested refusals of a minimal predicate, with the deterrence they produce treated as a swept parameter, and a random-audit defence against collusion (Sections V-H–V-I).
4) **A full working system.** Multi-agent engine, FastAPI console, PostgreSQL backend with database-enforced integrity, Docker deployment and a 205-function test suite (Sections IV–VI).
5) **A real-data evaluation methodology.** Per-pool Erlang-B calibration to two real metropolitan networks, cold-start validation, a data-integrity audit of the real reports, and paired tests with Holm correction and TOST equivalence (Section VIII).

E. Organisation

Section II reviews the literature. Section III states the research gaps, objectives, problem statement and project plan. Section IV gives the technical specification, Section V the design approach and formal model, and Section VI the design models (data-flow, use-case, class and sequence diagrams). Section VIII reports the evaluation, Section IX discusses implications and limitations, and Section X concludes.

II. Literature Review

Existing research is reviewed in five thematic areas that together define the problem MAHROS addresses, followed by a comparative analysis.

A. Inter-Hospital Transfer and Clinical Urgency

Clinical studies establish that escalation transfer is frequent, structured and time-critical. Iwashyna *et al.* mapped 47,820 Medicare critical-care transfers among 3,308 hospitals and found a stable core–periphery network in which more central hospitals had more capabilities [7]. That is the hub-and-spoke topology of the Houston region studied here, where the Texas Medical Center is surrounded by community hospitals. Churpek *et al.* associated each hour of delayed ICU transfer with about a 3% increase in adjusted odds of mortality [8], which motivates time to definitive care as the primary objective. Wong and Harris showed that inter-hospital critical-care transfer delays in 49 UK hospitals were driven by organisational rather than geographical factors [9]. Mueller *et al.* related transfer processes to patient outcomes [10]. Greenwood-Ericksen *et al.* provided the most direct evidence on refusals: in 26,020 requests, acceptance fell as the receiver's census and boarding rose, and overall acceptance dropped from above 70% before the pandemic to about 40% afterwards [14]. In the United States, the Emergency Medical Treatment and Labor Act already treats refusing a transfer despite available specialised capacity as unacceptable for emergency patients (42 CFR 489.24(f)) [2]. It does not cover patients who are already admitted, so it is a precedent for the accountability MAHROS provides, not a mechanism for it.

B. Centralised Coordination and Optimisation

Operational transfer centres coordinate placements centrally. The Arizona Surge Line coordinated COVID-19 transfers across public, private and federal hospitals statewide and reported equity of access as an outcome [11]. Optimisation models go further. Dijkstra *et al.* balanced COVID-19 patients

across Dutch hospitals and regions using occupancy forecasts and a queueing network with explicit fairness [12], and Baas *et al.* extended that line to stochastic programming of bed capacity over demand scenarios [52]. Parker *et al.* formulated linear and mixed-integer programmes, with robust variants, to redistribute demand and resources [51]. Mehrotra *et al.* optimised the sharing of a ventilator stockpile among states [13]. These approaches achieve high-quality joint allocations, including batched assignment by the Hungarian method [15], but all of them assume that the coordinator sees each hospital's true state. MAHROS adopts the strongest form of this approach, with optimisers that read the simulator's true private state, as its efficiency ceiling.

C. Queueing and Simulation Models of Hospital Capacity

Hospital bed pools are classically modelled as loss systems, in which a patient who arrives to a full unit is turned away or diverted. De Bruin *et al.* validated the Erlang loss model against several years of Dutch hospital data and built a decision-support tool for ward sizing. They showed that a single target occupancy across units is unrealistic, because small units must run emptier to keep refusal rates acceptable [16]. Green made the same point for U.S. hospitals: the widely used 85% occupancy target produces long delays in small units [17]. Little's law [18] links occupancy, arrival rate and length of stay in any stable system and underlies every calibration from census data. Discrete-event simulation is the standard method for evaluating hospital operations under such models [19]. MAHROS inverts the Erlang loss relation per bed pool to recover the arrival rates that reproduce each real hospital's reported occupancy, and then validates the result from a cold start.

D. Multi-Agent Negotiation and the Contract-Net Protocol

Multi-agent systems are well established in healthcare for scheduling, decision support and monitoring [20], [21], and discrete-event simulation is the standard tool for evaluating hospital operations [19]. The Contract-Net Protocol [22], standardised by FIPA [3], allocates a task by announcement, parallel bidding and award. The manager needs no knowledge of the contractors' internal state, which makes the protocol a natural fit for transfer brokering. Automated negotiation studies the protocols and decision models of self-interested agents more broadly [23]. Mechanism design seeks truthful reporting through incentives, from the sealed-bid second-price auction [24] to computational mechanisms [25]. Such mechanisms need payments or priced valuations, which public-interest bed allocation does not have. MAHROS therefore retains Contract Net's parallelism and privacy and replaces incentive compatibility with verification.

E. Argumentation and Formal Dispute Resolution

Dung's abstract argumentation frameworks [26] model a dispute as a graph of arguments and attacks. Of their semantics, the grounded semantics is unique, sceptical and computable in polynomial time [27], [53], and its labelling form and the reinstatement principle are analysed by Caminada [43]. Argumentation-based negotiation lets agents justify and criticise offers, not only make and reject them [28], and Amgoud *et al.* characterised when arguments change negotiation outcomes [44]. In medicine, argumentation supports clinical decision making [29], and the ProCLAIM model mediated deliberation over safety-critical actions in organ transplantation [30]. Python solvers such as PyArg are available [45]. MAHROS applies grounded semantics as an accountable resolution rule for evidence-backed disputes over refusals, so an outcome never depends on persuasion.

F. Trust, Reputation and Enforcement

The standard multi-agent answer to unreliable peers is trust and reputation. Agents learn whom to rely on from direct experience or third-party ratings, and surveys classify these models by information source, aggregation and robustness to manipulation [31], [32], [33]. Reputation works when interactions repeat and their outcomes are eventually observed. In transfer brokering the decisive outcome, whether the refuser actually had a bed, is never observed by anyone but the refuser, so the feedback loop reputation depends on is missing. Two other strands are more directly applicable. Economic analyses of enforcement show that random inspection combined with a credible sanction deters misconduct even at low inspection rates [34], which motivates the random spot checks in MAHROS. Empirical industrial organisation has detected bid rigging in procurement auctions from the joint pattern of bids [35], which motivates the co-refusal statistic used to monitor collusion.

G. Privacy-Preserving Multi-Agent Coordination

Distributed constraint optimisation (DCOP) is the formal setting closest to transfer brokering: agents with private constraints jointly choose an assignment. Léauté and Faltings showed how cryptographic techniques and distributed computation can protect four kinds of private information, namely constraint feasibility, decisions, agent identities and problem topology, with formally proven guarantees [36]. Greenstadt *et al.* measured empirically how much private information standard DCOP algorithms leak [46]. These approaches assume honest inputs and carry significant computational and engineering cost. MAHROS takes a lighter, measurable route: no occupancy value is ever transmitted, and the information each message does reveal is quantified as an entropy reduction (Section VII-D).

H. Tamper-Evident Records and Digital Signatures

Hash chains and Merkle trees make an append-only record tamper-evident [37], [47]. Schneier and Kelsey extended this to audit logs on machines that may later be compromised [38], and Crosby and Wallach introduced history trees that give logarithmic-size membership and consistency proofs [48]. Blockchain designs for health records add distributed consensus over who may write [49]. A consortium of identifiable hospitals needs tamper *evidence* rather than permissionless

consensus, which MAHROS provides with a Merkle-linked ledger and a trigger-maintained audit chain with external checkpoints. For signed refusals, Ed25519 provides fast, compact, non-repudiable signatures with 32-byte public keys and 64-byte signatures [39].

I. Comparative Analysis of Existing Approaches

Table I compares MAHROS with representative approaches along the architectural axes that matter for transfer brokering.

III. Research Gap, Objectives and Project Plan

A. Research Gap

The review reveals four gaps that current transfer-coordination approaches leave open.

1) **The disclosure dilemma.** Parallel, high-quality allocation is achieved today only by centralising every hospital's live occupancy in a registry or optimiser [11], [12], [51]. This creates a sensitive data monopoly, and it moves the incentive to misreport one layer down into the reported data instead of removing it. Decentralised negotiation keeps occupancy private but returns sealed refusals that cannot be checked [22].
2) **No evidence base for private claims.** Argumentation-based negotiation specifies how agents exchange justifications [28], [44], but not what a justification about a private quantity, a hospital's free beds, can be checked against without disclosing that quantity. Reputation systems cannot fill the gap, because whether a refuser truly had a bed is never observed [31].
3) **Misreporting is outside the evaluation.** Coordination studies evaluate allocation quality under truthful reporting, and negotiation studies use synthetic tasks. No transfer-brokering mechanism has been evaluated under parametrised strategic refusal on a real hospital network, with a control for the risk of wrongly accusing truthful but cautious hospitals.
4) **No inspectable accountability layer.** Existing systems keep no tamper-evident record linking each award, refusal and challenge to the evidence used, so a regulator cannot audit why a patient went where they did, and a hospital cannot be held to what it said.

B. Objectives

MAHROS is designed around seven measurable objectives. Each is verified in Section VIII.

- O1 **Parallel negotiation.** Decide each round within a single 2-minute bid window, and place at least 5 percentage points more patients within their clinical window than a perfect-information telephone search on the real Houston surge week.
- O2 **Near-optimal efficiency without disclosure.** Achieve a success rate statistically equivalent, within a pre-declared ± 2-point margin (TOST, $\alpha = 0.05$), to optimisers that read every hospital's true occupancy.
- O3 **Verifiable refusals.** With half the hospitals refusing falsely, recover at least 50% of the lost placements while wrongly overruling *zero* truthful hospitals.
- O4 **Privacy.** Transmit no occupancy count in any message, and ensure no node receives more than a small share of capacity-bearing messages (decision concentration far below the 1.0 of a central node).
- O5 **Tamper evidence.** Lock all real reference data against application and owner edits. Record every change to live state and records in a hash-chained audit log in which any edited, inserted or deleted row is detected.
- O6 **Realism.** Reproduce the occupancy that real hospitals reported, from an empty start, with a per-facility mean absolute error below 5 points in the reference week of both cities, the second without re-tuning.
- O7 **Responsiveness.** Keep the 99th-percentile protocol decision time per transfer below 1 s, so that computation never consumes clinical time.

C. Problem Statement

> *To design, implement and empirically validate a decentralised multi-agent system that brokers inter-hospital escalation transfers. In this system each hospital's agent answers from private state, every refusal is verifiable against public evidence without disclosing occupancy, disputes are resolved by a formal and accountable rule, and every decision is recorded tamper-evidently. It is evaluated on real hospital networks against current practice and full-information optimisers.*

Formally, let $\mathcal{H}$ be the hospitals and $\mathcal{R} = \{\text{WARD}, \text{HDU}, \text{ICU}, \text{OR}, \text{VENT}, \text{CATH}\}$ the bed-pool types. Hospital h has capacity $c_{h,r}$ and private occupancy $x_{h,r}(t)$. A request $q = (o_q, r_q, s_q, a_q, t_q, \ell_q)$ carries origin, pool, specialty, acuity $a_q \in \{1, \dots, 5\}$, creation time and expected stay. Its clinical window $W(a_q)$ (720, 480, 240, 150 and 90 min for acuity 1–5) sets the deadline $d_q = t_q + W(a_q)$. Assigning q at time t to h starts care at $T_q(h) = t + \tau_{o_q,h} + p_h(q)$, the sum of road time including handover and receiver preparation. The objective is lexicographic:

$$\text{(i)} \quad \max \textstyle\sum_q \mathbf{1}[T_q(h_q) \le d_q], \tag{1}$$

$$\text{(ii)} \quad \min \textstyle\sum_q (T_q(h_q) - t_q), \tag{2}$$

$$\text{(iii)} \quad \min G(\{b_h/\sigma_h\}_{h \in \mathcal{H}}), \tag{3}$$

where b_h counts acceptances by h, σ_h is its share of network capacity and G is the Gini coefficient. These are subject to capability and capacity feasibility, a *privacy* constraint (no message contains $x_{h,r}$) and a *self-interest* condition (a hospital may report a refusal its state does not support).

D. Project Plan

1) Work Breakdown Structure: The implementation was organised into six work packages (WPs).

TABLE I
COMPARATIVE ANALYSIS OF INTER-HOSPITAL TRANSFER COORDINATION APPROACHES

Feature	Telephone search [9]	Transfer centre / registry [11]	Central optimiser [12], [51]	Contract Net [22]	Argumentation-based negotiation [28], [44]	**MAHROS (this work)**
Topology	Serial, peer-to-peer	Centralised	Centralised	Decentralised	Decentralised	Decentralised multi-agent
Search	One hospital at a time	Parallel lookup	Joint solve	Parallel bids	Parallel, iterative	Parallel bids, 2 rounds
Occupancy disclosed	Verbally, per call	Every hospital, continuously	Every hospital, continuously	No	Partially	**Never**
Refusal verified	No	No	No (assumes truth)	No	Arguments, no evidence base	**Registry + ledger evidence**
Dispute resolution	None	None	None	None	Persuasion / utility	**Grounded semantics**
Misreporting modelled	–	–	No	No	Partly	**Yes, swept**
Collusion defence	–	–	–	–	–	**Random spot checks**
Audit trail	Notes	Database	–	–	–	**Hash-chained ledger + audit log**
Evaluation data	Clinical cohorts	Operational	Real/retrospective	Synthetic	Synthetic	**57 real hospitals**

- **WP1: Core engine.** Resource and hospital agents, Contract-Net protocol, message bus, discrete-event simulator, live console.
- **WP2: Deliberation and evaluation tooling.** Argumentation layer, challenge engine, HHS calibration, paired-test and TOST statistics.
- **WP3: Deployment.** Docker packaging, deployment configuration and seed data.
- **WP4: Database.** PostgreSQL schema as the source of truth, trigger locks, hash-chained audit log and the Hospital-data tab.
- **WP5: Real-data evaluation.** Warm start, California case-mix, cold-start validation, Phoenix region, data-integrity audit, registry, collusion and reputation experiments.
- **WP6: Hardening and reporting.** Reproducibility fixes, indexed ledger queries, generated results document and this paper.

2) Project Schedule: Figure 1 shows the schedule, derived from the repository's commit history (first to last commit of each package, rounded to whole weeks, weeks starting Monday 10 August 2026).

3) Milestones and Verification Gates: Each milestone was accepted against a verification gate expressed as passing tests or a generated result (Table II).

4) Risk Management: Table III lists the principal project risks and the mitigations that were actually implemented.

TABLE II
MILESTONES AND VERIFICATION GATES

ID	Milestone	Date	Verification gate
M1	Protocol prototype	15–16 Aug	Contract Net places patients; live console streams every step; same seed gives same result
M2	Deliberation layer	27 Aug	Grounded labelling passes textbook cases (reinstatement, even cycle); honest hospitals never overruled
M3	Deployable system	3–9 Sep	Docker stack starts in one command; dashboard test count matches suite
M4	Database of record	13 Oct	App cannot change capacity; owner cannot change real data; edited or deleted audit row breaks the chain
M5	Real-data evaluation	13–14 Oct	Cold-start occupancy error below 5 pts; Phoenix replicates with unchanged code
M6	Final results	15–20 Oct	All results regenerated from JSON; risk-based checks reported as negative

TABLE III
RISK IDENTIFICATION AND MITIGATION MATRIX

ID	Risk	Severity	Implemented mitigation
R1	Results depend on simulated behaviour	High	Every behavioural parameter swept; truthful-but-cautious control arm; ranges reported
R2	Simulated hospitals unlike real ones	High	Per-pool Erlang-B fit to HHS data; cold-start validation; untuned second city
R3	Non-reproducible runs	High	SHA-256-seeded random streams; fresh-process reproducibility test (two defects found and fixed)
R4	Truthful hospitals wrongly penalised	Critical	Two-predicate attestation; spot checks restricted to capacity claims; safety tests
R5	Collusion defeats reactive checks	High	Uniform random spot checks; cartel experiments
R6	Tampering with records	High	Trigger locks, column grants, hash-chained audit log, checkpoints
R7	Deliberation cost grows with run length	Medium	Indexed ledger queries (5× faster audit, identical output)

IV. TECHNICAL SPECIFICATION

A. Functional Requirements

The requirements are organised by pipeline stage. Each is implemented and covered by the test suite.

Stage 1: Network and data ingestion (FR-1). FR-1.1: build each study network from the HHS facility reports [4] for a chosen week: staffed inpatient beds, occupied beds, staffed adult ICU beds and occupied ICU beds, keyed by CMS Certification Number (CCN). FR-1.2: join capability (cardiac, neurological, trauma, obstetric, burns, catheterisation-room count) from the CMS Provider of Services file [5] on CCN. FR-1.3: load pairwise OSRM road times [50] and add a handover allowance. FR-1.4: label every simulator input as

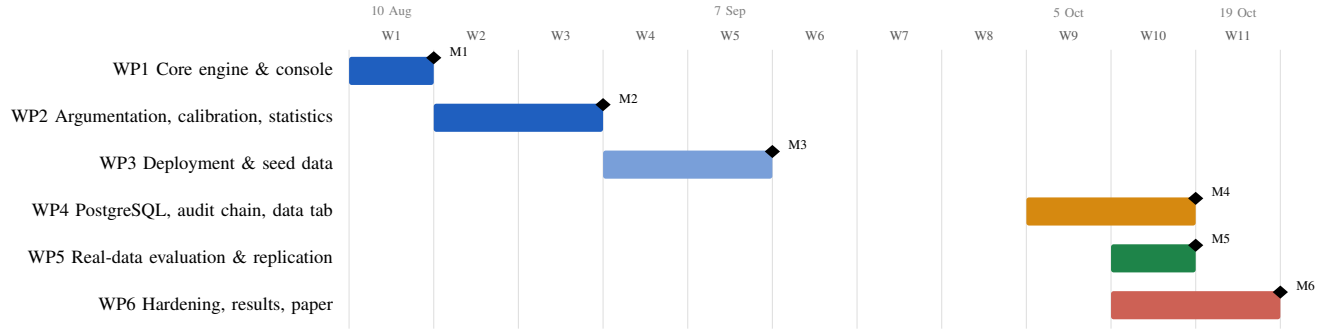


Fig. 1. Project schedule (11 weeks, 10 August to 25 October 2026) derived from the commit history; diamonds mark milestones M1–M6 (Table II). Weeks 6–8 contain no commits.

observed, derived or assumed, and generate the provenance report from code.

Stage 2: Agent assessment (FR-2). FR-2.1: each resource agent assesses a request from its own pool state and returns feasibility, preparation time, capability match $\kappa \in [0, 1]$, post-acceptance strain and a private opportunity cost. FR-2.2: a hospital's bidding policy decides what to report. The truthful assessment is retained privately for defence. FR-2.3: the hospital's self-protection reserve is relaxed for acuity 3 and released entirely for acuity ≥ 4.

Stage 3: Negotiation (FR-3). FR-3.1: announce an anonymised call for proposals (pseudonymous patient token, resource, specialty, acuity, deadline, expected stay) to the eight nearest peers by road time, and to sixteen in a second round. FR-3.2: collect all bids in one parallel 2-minute window. FR-3.3: reserve the winner's unit for 45 min, and re-award to the runner-up if the reservation race is lost. FR-3.4: retry failed requests every 20 min while the deadline allows.

Stage 4: Verification and deliberation (FR-4). FR-4.1: append every capacity refusal to the ledger before examining it. FR-4.2: challenge a capability refusal that the public register contradicts. FR-4.3: challenge a capacity refusal when the refuser accepted the same resource within 60 min, or has a recent refusal-then-acceptance history. FR-4.4: let the challenged hospital defend or sign, or have its refusal struck out by grounded labelling, and hold it to its truthful bid. FR-4.5: spot-check a configurable fraction of uncontradicted capacity refusals. FR-4.6: never compel a hospital beyond its real capacity.

Stage 5: Award and fairness (FR-5). FR-5.1: discard bids that cannot meet the deadline before scoring. FR-5.2: score surviving bids on time to care, capability, strain and fairness credit, collapsing towards speed for acuity ≥ 4. FR-5.3: allow a burden objection only for acuity ≤ 3 and only when a comparable alternative exists. FR-5.4: attach a plain-English rationale to every award.

Stage 6: Recording and audit (FR-6). FR-6.1: append each agreement, with the hash of its canonical terms, to a Merkle-linked ledger block. FR-6.2: persist each negotiation (request, every agent decision, challenges, allocation, ledger block) in one database transaction. FR-6.3: audit every insert, update and delete with before and after values and actor. FR-6.4: provide chain verification and checkpointing.

TABLE IV
QUANTITATIVE NON-FUNCTIONAL REQUIREMENTS AND VERIFICATION

ID	Category	Requirement	Verification
NFR-1	Latency	p99 decision time < 1 s per transfer	`realtime.py` timing of every transfer
NFR-2	Throughput	one bid window per round, independent of peer count	protocol design; message bus log
NFR-3	Privacy	no occupancy value in any message; no global view	bus audit; leakage model; tests
NFR-4	Safety	zero false accusations of truthful hospitals	cautious-hospital control arms; tests
NFR-5	Integrity	real data immutable; any audit-row edit detected	database tests as owner and app role
NFR-6	Atomicity	a negotiation is fully persisted or not at all	single-transaction write; smoke test
NFR-7	Reproducibility	identical results for (scenario, seed) in any process	fresh-process test
NFR-8	Realism	cold-start occupancy MAE < 5 pts	cold-start validation script
NFR-9	Portability	core runs on the Python standard library	dependency-free install; CI-style test run
NFR-10	Determinism	award never delegated to a language model	LLM off by default; ablation flag

Stage 7: Monitoring and interaction (FR-7). FR-7.1: stream every protocol step to the console over WebSocket. FR-7.2: let a reviewer edit occupancy and hospital behaviour, then restore the reported data. FR-7.3: flag impossible or copied weekly reports (Data Integrity agent). FR-7.4: report unmet needs and the fullest pools (Regional Monitor).

B. Non-Functional Requirements

Table IV lists the quantitative non-functional requirements and how each is verified.

C. Assumptions and Constraints

Assumptions. (A1) Each hospital's capability (specialties, catheterisation rooms) is public and accurately listed in the CMS register. (A2) Agreements are recorded on a ledger that no single hospital can rewrite. (A3) Hospitals may misreport their own capacity but cannot forge other hospitals' ledger entries or the independent co-signature. (A4) Road time plus a fixed handover allowance is common knowledge.

Constraints. (C1) No message may contain a hospital's occupancy count. (C2) A patient may be assigned only to a

hospital that can treat them, has a unit free above its reserve, and can begin care before the deadline. (C3) Fairness may reorder clinically comparable options but may never delay a critically ill patient. (C4) No decision may depend on a non-deterministic component; the language model, if enabled, only explains.

D. Feasibility Study

1) Technical feasibility: The protocol's computational load is small. A negotiation evaluates at most sixteen peers, builds an argumentation framework of a few dozen arguments, and computes its grounded labelling by fixed-point iteration in time polynomial in the number of arguments [53]. Ed25519 signing and verification take about 0.43 ms and 0.86 ms per operation [39]. Ledger lookups are indexed by receiver, refuser and time. The measured median decision time is 3.55 ms with ledger-backed challenge and 13.9 ms in the adversarial setting (Section VIII-K), several orders of magnitude below the 90-minute shortest clinical window. A full 14-day simulation of 27 hospitals runs on a laptop.

2) Data feasibility: Every input that decides *who can take a patient* is public. Beds, ICU beds and weekly occupancy come from HHS [4], capability and catheterisation rooms from CMS [5], roads from OpenStreetMap through OSRM [50], and case-mix from California HCAI [6]. All are public domain or openly licensed and are vendored in the repository, so the system runs offline. For Houston, 8 inputs are observed, 4 derived and 6 assumed (Table VIII).

3) Economic and operational feasibility: All software is open source: Python, FastAPI, PostgreSQL, pgAdmin and Docker. The system runs on commodity hardware with no licence or cloud cost. Operationally, MAHROS reduces coordinator time per transfer from 50.3 min (telephone search) to 1.1 min (Section VIII-E). A deployment would integrate with each hospital's existing bed-management system instead of replacing it.

4) Social, legal and ethical feasibility: No patient-identifiable data are processed. Patients are referenced only by HMAC-SHA256 pseudonyms, and announcements carry clinical need but no demographic attributes. Hospitals never disclose occupancy, which removes the main commercial obstacle to regional data sharing. The accountability MAHROS adds mirrors an obligation regulators already recognise for emergency patients [2].

E. System Specification

1) Hardware: MAHROS has no special hardware requirements. Table V gives the host on which the Docker deployment and live console were run for this paper.

2) Software stack: Table VI lists the software stack, with versions read from the running deployment.

TABLE V
DEVELOPMENT AND DEPLOYMENT HOST

Component	Specification
Processor	Intel Core i5-13420H (8 cores, 12 threads)
Memory	16 GB (15.6 GB usable)
Operating system	Windows 11; Docker Desktop with Linux containers
GPU	Not required
Network	Not required at run time (data vendored)

TABLE VI
SOFTWARE STACK (VERSIONS FROM THE RUNNING DEPLOYMENT)

Layer	Component	Version / configuration
Runtime	CPython	3.13.17 (`python:3.13-slim`); core uses stdlib only
Web API	FastAPI / Starlette	0.162.0 / 1.9.0, served by Uvicorn 0.54.0
Live channel	WebSocket	protocol events streamed per step
Database	PostgreSQL	16.15 (`postgres:16-alpine`)
DB driver	psycopg	3.3.6 (binary)
Signatures	cryptography	50.0.2, Ed25519; HMAC-SHA256 fallback
Admin UI	pgAdmin 4	8.14
Orchestration	Docker / Compose	29.3.1 / v5.1.1, three services
Analysis	stdlib statistics	paired t, TOST, Holm, bootstrap
Testing	pytest	205 test functions; DB tests with live Postgres
Optional	Ollama / hosted LLM	explanations only; off by default

V. DESIGN APPROACH AND DETAILS

A. System Architecture

Figure 2 shows the layered architecture. Real public data are calibrated into a network of hospital agents. A transfer manager per request runs the negotiation, a Verifier checks refusals against the public register and the transfer ledger, and every agreement, refusal and agent decision is persisted in PostgreSQL behind a hash-chained audit log. The same engine runs in the batch simulator and in the live console.

B. Mathematical Formulation of the Pipeline

Each transfer request q is processed by a sequence of deterministic transformations, one per protocol phase.

1) *Announcement.* The origin selects the peer set by public road time only:
$$P_q^{(k)} = \arg\min^{(8k)}_{h \in \mathcal{H} \setminus \{o_q\}} \tau_{o_q,h}, \qquad k \in \{1,2\}, \quad (4)$$
the $8k$ nearest hospitals in round k, and sends each the public view $\pi(q)$, which carries no patient identity and no demographic data.
2) *Private assessment and report.* Each peer computes a truthful assessment $A_h(q) = \mathrm{Assess}_h(\pi(q), x_h)$ and reports $\hat{A}_h(q) = \beta_h(A_h(q), q)$ through its bidding policy β_h. Only $\hat{A}_h$ leaves the hospital.
3) *Verification.* For each refusal the Verifier computes a challenge set from public evidence only, namely the capability register $\mathcal{G}$ and the ledger L:
$$C_q = \{\mathrm{ch}(h) : \hat{A}_h(q) = \textsc{ref} \wedge \mathrm{Contra}(h, q, \mathcal{G}, L, t)\}, \quad (5)$$
where REF denotes a refusal.
4) *Deliberation.* Bids, refusals, challenges and defences form an argumentation framework $\mathcal{AF}_q$, whose grounded labelling $\lambda_q = \mathrm{Grd}(\mathcal{AF}_q)$ strikes out undefended refusals. Each struck-out refusal is replaced by the hospital's truthful bid.

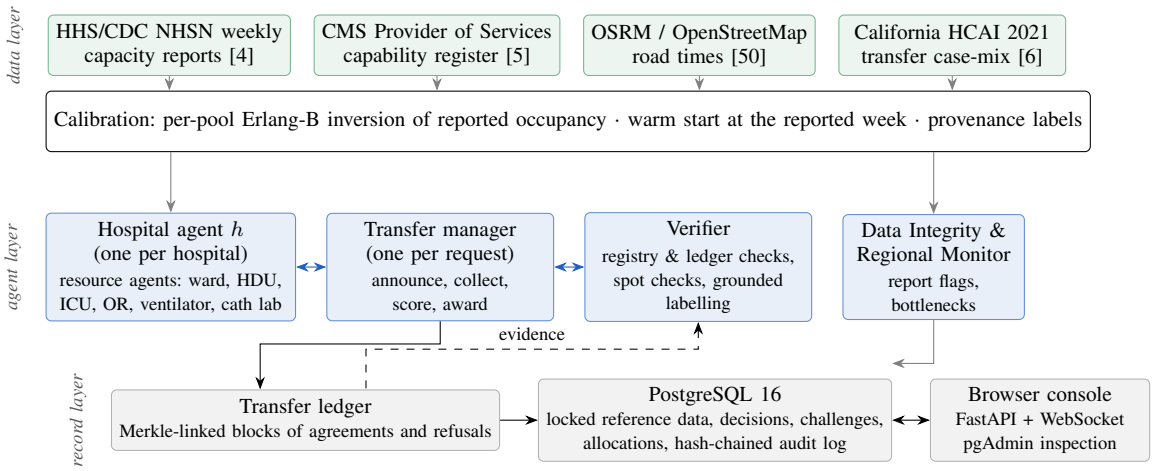


Fig. 2. System architecture of MAHROS: data layer (real public sources), agent layer (hospital, transfer-manager, verifier, data-integrity and regional-monitor agents) and record layer (ledger, PostgreSQL, console).

TABLE VII
STUDY NETWORKS BY TIER, REFERENCE WEEK 10 JANUARY 2021 (REPORTED DATA)

Region	Tier	Hosp.	Inpatient beds	ICU beds	Cath lab	Mean occ.
Houston	1 (community)	10	675	114	8	0.83
	2 (district)	11	2,890	260	10	0.93
	3 (tertiary)	6	4,611	656	6	0.81
Phoenix	1 (community)	11	896	119	5	0.56
	2 (district)	8	2,356	382	6	0.82
	3 (tertiary)	11	6,220	959	8	0.77

5) *Award.* Among the surviving feasible bids $F_q = \{h : T_q(h) \le d_q\}$, the winner is

$$h_q^\star = \arg\max_{h \in F_q} \big(S_h,\ -T_q(h),\ -\mathrm{id}(h)\big) \text{ (lexicographic)}, \tag{6}$$

where S_h is the score of Eq. (15).

6) *Record.* The agreement and every capacity refusal are appended to the ledger, $L \leftarrow L \,\|\, \langle q, h_q^\star, \mathrm{H}(\text{terms})\rangle$, and persisted atomically with the audit chain of Eq. (17).

C. Data and Calibration Layer

Study networks. The headline network contains the 27 short-term acute hospitals with at least 25 inpatient beds within 40 km of the Texas Medical Center in Houston: 8,176 inpatient beds and 1,029 staffed adult ICU beds in the reference week of 10 January 2021. Two multi-site system aggregates reporting hundreds of beds at administrative addresses are excluded, with stated reasons. The replication network contains 30 hospitals in metropolitan Phoenix, selected by the same code. Scenarios are real weeks chosen by observed occupancy (Table XIV). Houston ran fuller in December 2023 than during the COVID-19 waves.

Table VII summarises both networks by tier in their reference week.

Erlang-B calibration. Each ward, HDU and ICU pool is modelled as an Erlang loss system with c beds and offered load a [16], [17]. The blocking probability follows the stable recursion

$$B(k,a) = \frac{a\,B(k-1,a)}{k + a\,B(k-1,a)}, \quad k = 1,\dots,c, \quad B(0,a) = 1, \tag{7}$$

and for the reported occupancy ρ_h the offered load is obtained by bisection on

$$a_{h,r}\big(1 - B(c_{h,r}, a_{h,r})\big) = \min(\rho_h, \bar\rho)\, c_{h,r}, \qquad \bar\rho = 0.95. \tag{8}$$

By Little's law [18], the pool's arrival rate is $\lambda_{h,r} = a_{h,r}/\bar W_r$, using the realised lognormal mean stay ($\sigma = 0.6$), scaled to a network mean of 4.6 days. The fit is per pool rather than aggregate, so an ICU-heavy facility does not overflow its ward. The cap $\bar\rho$ exists because a stationary loss system cannot sit at 100% occupancy; facilities above the cap are fitted to it and keep their reported value for validation.

Warm start and escalation. Runs start at the occupancy reported that week. Patients already present receive a length-biased remaining stay: for a lognormal(μ,σ) stay, the length-biased stay is lognormal$(\mu+\sigma^2,\sigma)$, and the time left is a uniform fraction of it. Each admission escalates with probability 0.14 after a lognormal delay (mean 7 h), which gives transfer requests for about 4.5% of admissions, consistent with the U.S. critical-care transfer share [7]. The resource each request needs follows the case-mix of 37,439 real California transfers (Table IX).

Provenance. Table VIII lists every simulator input with its provenance class.

D. Agent Layer

Resource agents. One agent per pool assesses a request. It refuses with `resource_not_offered` if the pool has zero capacity, with `no_specialty_capability` if the capability match is $\kappa = 0$, and with `at_self_protection_reserve` if the free units do not exceed the reserve. The match κ is 1 for a listed specialty, 0.9 for general care and 0.6 for an unlisted specialty at a tertiary centre. The reserve is one ICU unit and two

TABLE VIII
PROVENANCE OF SIMULATOR INPUTS (HOUSTON): 8 OBSERVED, 4 DERIVED, 6 ASSUMED

Input	Class	Source or rule
Specialty capability	observed	CMS POS Q1 2021 certification codes
Catheterisation labs	observed	CMS POS catheterisation room count
ICU beds	observed	HHS staffed adult ICU beds (7-day avg.)
Paediatric capability	observed	HHS staffed paediatric ICU beds > 0
Identifier, name, coordinates	observed	HHS CCN, name, geocoded address
Travel time	observed	OSRM over OpenStreetMap
Ward beds	derived	inpatient beds minus ICU and HDU
Admission mix	derived	observed bed composition / pool stay
Arrival rate	derived	Erlang-B inversion, Eq. (8)
Tier	derived	beds vs. national percentiles, gated on ICU
HDU beds; ventilators	assumed	$0.5\times$ and $0.8\times$ ICU beds
OR slots; coordinators	assumed	1 per 30 and 1 per 150 inpatient beds
Handover; mean stay	assumed	12 min (swept 8–20); 4.6 days (cited)

TABLE IX
ESCALATION CASE-MIX FROM 37,439 CALIFORNIA TRANSFERS (2021)

Profile	Share	Profile	Share
HDU, general	19.7%	ICU, neuro	9.1%
ICU, general	18.2%	OR, general	5.7%
Ventilator	15.1%	HDU, obstetric	3.5%
Cath lab, cardiac	12.5%	OR, trauma	2.5%
ICU, cardiac	12.5%	ICU, paediatric	1.1%
		ICU, burns	0.1%

ward units, falls by one for acuity 3, and is zero for acuity ≥ 4. Otherwise the agent reports preparation time, κ, and the post-acceptance strain

$$\sigma^{+}_{h,r}(q) = \frac{x_{h,r} + y_{h,r} + 1}{c_{h,r}}, \tag{9}$$

where $y_{h,r}$ counts reservations. It also computes a private opportunity cost $\pi_h(q) = (\sigma^{+})^3(1 + 0.25\,\ell_q/1440)$, which never leaves the hospital.

Hospital agents and behaviour policies. A bidding policy maps the truthful assessment to the reported one. HONEST reports the truth. PROTECTIVE fabricates a refusal with probability ϕ when the patient is expensive ($\ell_q \geq 24\,\text{h}$) or the hospital would exceed a comfort threshold θ (default 0.55), never for acuity 5, and chooses the most plausible excuse. CAUTIOUS never lies but holds two extra units in reserve; it is the control against false accusations. COLLUDING is protective and also refuses whenever a fellow cartel member refused the same resource in the last 60 min. Policies are assigned in nested, incentive-ordered fashion, starting from tertiary centres.

E. Contract-Net Negotiation

Algorithm 1 gives the negotiation. Its distinctive step is deliberation (lines 5–15) between bidding and award. All bids arrive in one parallel window, so a round costs 2 min regardless of the number of peers; that parallelism is what replaces the serial telephone search.

Algorithm 1 MAHROS negotiation for request q at time t

Input: request q, hospitals $\mathcal{H}$, ledger L, register $\mathcal{G}$
Output: agreement or failure

1: **for** $k = 1, 2$ **do**
2: $P \leftarrow$ the $8k$ peers nearest to o_q (Eq. 4)
3: send CFP($\pi(q)$) to P; $B \leftarrow \{\hat{A}_p(q)\}_{p \in P}$
4: $\mathcal{A} \leftarrow$ bid and refusal arguments for B
5: L.APPENDREFUSAL(b) for every capacity refusal b
6: **for all** refusals $b \in B$ **do**
7: **if** attested, capacity reason, $U < \theta_{\text{spot}}$ **then**
8: spot check: ATTEST(b) or challenge
9: **else if** CONTRA($b, \mathcal{G}, L, t$) **then**
10: challenge; defence if DEFEND(b) (or ATTEST)
11: **end if**
12: **end for**
13: add admissible burden objections
14: $\lambda \leftarrow$ GROUNDED($\mathcal{A}$)
15: replace each OUT refusal by its truthful bid
16: $t \leftarrow t + 0.75$; rank on-time bids by Eq. (15)
17: **for** candidate h in rank order **do**
18: **if** h.RESERVE($q, t + 45$) **then**
19: L.APPENDAGREEMENT(q, h); **return** agreement
20: **end if**
21: **end for**
22: **if** $t + 2 \geq d_q$ **then break**
23: **end if**
24: **end for**
25: **return** failure

F. Refusal Verification

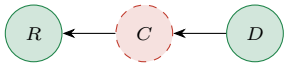
The design principle is that *capability is public and capacity is private*. A capability refusal is checked against the public register. Reason `no_specialty_capability` is contradicted if $s_q \in \mathcal{G}_h$, and `resource_not_offered` if h is listed as operating pool r_q with positive capacity. A capacity refusal is checked against the refuser's own recorded commitments, the only public evidence that bears on it. It is directly contradicted if

$$\exists\, e \in L:\ e.\text{receiver} = h,\ e.\text{resource} = r_q,\ t - 60 \leq e.t \leq t, \tag{10}$$

that is, if the hospital accepted a patient needing the same resource within the hour. Otherwise it is challenged on history if, within the last 240 min, the hospital made a recorded capacity refusal followed within 60 min by its own acceptance of the same resource. Every capacity refusal is appended to the ledger *before* examination, so it becomes evidence against later ones. The check achieves accountability without disclosure: it never reads a hospital's state, only what the hospital publicly committed to.

G. Dispute Resolution by Grounded Semantics

Each negotiation builds a Dung framework $\mathcal{AF} = (\mathcal{A}, \rightarrow)$ [26] with exactly four attack rules. A challenge attacks the refusal it names. A defence or attestation attacks the challenge it answers. A burden objection attacks the objector's own bid. A counter-offer attacks the hospital's original bid. Hospitals never attack one another directly. The labelling $\lambda : \mathcal{A} \rightarrow$



(a) truthful: refusal reinstated



(b) fabricated: refusal struck out

Fig. 3. Grounded labelling of a challenged refusal. R refusal, C evidence-backed challenge, D defence or signed attestation. Green: IN; red dashed: OUT.

$\{\text{IN}, \text{OUT}, \text{UNDEC}\}$ is the least fixed point of

$$\lambda(A) = \text{IN} \;\Leftarrow\; \forall B \to A:\ \lambda(B) = \text{OUT}, \tag{11}$$

$$\lambda(A) = \text{OUT} \;\Leftarrow\; \exists B \to A:\ \lambda(B) = \text{IN}, \tag{12}$$

which yields the unique grounded extension [27]. Figure 3 shows the two patterns that arise: reinstatement [43] when the hospital defends, and defeat when it cannot. A struck-out refusal is replaced by the hospital's truthful bid. If that bid is still infeasible, the refusal stands, so no hospital is ever compelled beyond its real capacity. Because an honest refusal is backed by the refuser's real state, it can always be defended, and truthful hospitals are never overruled.

H. Signed Refusals (Attestation)

In the attested protocol a challenged hospital discharges the challenge only by signing the contested predicate. It may sign NO_UNIT_AVAILABLE (no unit was free for this request at time t) or AT_DECLARED_RESERVE (the only free units lie inside a declared reserve). The second predicate lets a cautious hospital certify its true position. The canonical payload (hospital, request, resource, predicate, time, nonce) is signed with the hospital's Ed25519 key [39] and co-signed by an independent key, such as a regulator or record-system vendor, so a cartel cannot vouch for itself. No bed count is ever signed. A protective hospital that fabricated a refusal declines to sign with probability δ, the deterrence, and otherwise signs falsely. Post hoc, a signed "no unit available" followed within 5 min by the same hospital's acceptance of that resource generates an advisory *audit referral* for a regulator.

I. Collusion Defence

A cartel whose members never accept a resource while any member's refusal of it is outstanding leaves no contradiction for Eq. (10) to find. MAHROS therefore adds two mechanisms. The first is *co-refusal monitoring*, which tracks the temporal follow-on lift

$$\text{FL}(a, b) = \frac{\Pr[b \text{ refuses } r \mid a \text{ refused } r \text{ within } 60\,\text{min}]}{\Pr[b \text{ refuses}]} \tag{13}$$

as an advisory signal [35]. The second is *random spot checks*: with probability θ_{spot} a capacity refusal that nothing contradicts is challenged anyway and must be signed. This is random inspection [34] in protocol form. A risk-based variant checks hospitals already caught at a higher rate $\theta' $.

J. Award Scoring and Fairness

Bids that cannot meet the deadline are discarded, never traded off. For each surviving bid the time term blends relative speed with the absolute safety margin:

$$\tau_h^\star = 0.6\Big(1 - \tfrac{T_h - T_{\min}}{T_{\max} - T_{\min}}\Big) + 0.4\,\text{clip}_{[0,1]}\Big(\tfrac{d_q - T_h}{d_q - t_q}\Big). \tag{14}$$

The score is

$$S_h = 0.45\,\tau_h^\star + 0.25\,\kappa_h + 0.15\,(1 - \sigma_h^+) - 0.15\,\gamma_h, \tag{15}$$

and for acuity ≥ 4 it collapses to $S'_h = 0.75\,(0.45\tau_h^\star + 0.25\kappa_h)/0.70 + 0.25\,S_h$, so equity never costs a critically ill patient time. The fairness credit is the relative deviation of a hospital's recent acceptances from its capacity share over the last 200 acceptances:

$$\gamma_h = \text{clip}_{[-1,1]}\left(\frac{n_h/n - \sigma_h}{\sigma_h}\right). \tag{16}$$

A hospital with $\gamma_h > 0.35$ may object to its own bid, but only for acuity ≤ 3 and only when another bid is within 15 min of its own time to care.

K. Ledger and Database Integrity

Agreements and capacity refusals form a ledger of Merkle-linked blocks of 16 entries [37], [47]. Each block header commits to the previous block's hash and the Merkle root of its entries, and each agreement carries the SHA-256 hash of its canonical terms. Per-receiver, per-refuser and time-window indices keep every challenge lookup sub-linear. In PostgreSQL, real reference data (hospitals, capabilities, weekly reports, travel times) have no application write grant and are locked by a `BEFORE` trigger, even against the owner. Live state (occupancy, behaviour policy) is editable only through column grants and a capacity `CHECK`. Records are append-only. Every change fires a trigger that appends to a hash-chained audit log under an advisory lock:

$$\begin{aligned} h_i &= \text{SHA256}(m_i), \\ m_i &= h_{i-1} \,\|\, ts_i \,\|\, actor_i \,\|\, op_i \,\|\, table_i \\ &\quad \|\, pk_i \,\|\, before_i \,\|\, after_i, \end{aligned} \tag{17}$$

with $h_0 = 0^{64}$. A verification function recomputes the chain and reports the first broken row, which catches any edited, inserted or deleted audit row, together with any checkpoint that no longer matches. A checkpoint exported outside the database also detects a wholesale rewrite by a superuser.

L. Request Lifecycle

Figure 4 gives the state machine of a transfer request as implemented.

VI. Design Models and Diagrams

A. Data Flow Diagrams

1) Level 0 (context): Figure 5 shows MAHROS as a single process exchanging data with its four external entities.

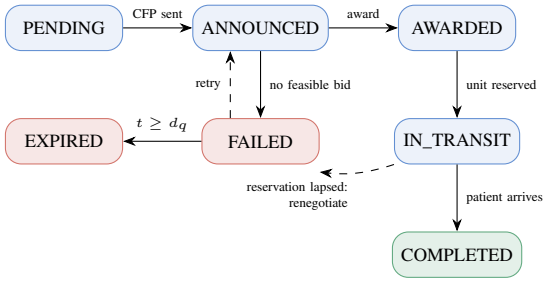


Fig. 4. Transfer-request state machine. A request completes only if care starts before its deadline d_q; otherwise it expires.

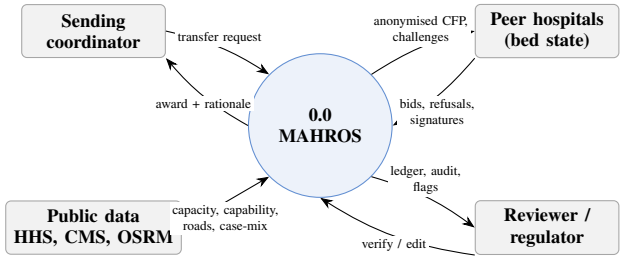


Fig. 5. Level 0 data flow diagram (context level).

2) Level 1 (subsystem decomposition): Figure 6 decomposes the system into seven processes and four data stores. Hospital private state (D4) is read only by process 3.0, running inside each hospital's own agent.

3) Level 2 (process 4.0, verify and deliberate): Figure 7 unpacks the deliberation process, the core of the verification mechanism.

B. Use-Case Model

Figure 8 gives the use-case diagram. Three use cases are specified below.

UC1: Request transfer. Actor: sending coordinator. Precondition: the patient is admitted and needs a resource unavailable locally. Main flow: (1) the coordinator selects resource, specialty and acuity; (2) the transfer manager pseudonymises the patient and announces a CFP to the eight nearest peers; (3) bids are collected in one window; (4) deliberation and award run (UC3–UC5); (5) the coordinator receives the receiving hospital, expected time to care and rationale. Post-condition: the agreement is on the ledger and the request, decisions and allocation are in PostgreSQL. Alternative flow: if no feasible bid survives and time remains, a second round asks sixteen peers; otherwise the request is retried every 20 min until its deadline.

UC3/UC4: Challenge and defend a refusal. Actors: Verifier, hospital agent. Trigger: a refusal is contradicted by the register or the ledger, or is spot-checked. Main flow: (1) the Verifier adds a challenge citing the ledger entries or register line; (2) the hospital defends or, in the attested protocol, signs the contested predicate; (3) grounded labelling reinstates the refusal. Alternative flow: if the hospital cannot defend or declines to sign, the refusal is struck out and the hospital is held to its truthful bid. If that bid is also infeasible, the refusal stands.

UC2: Bid on a request. Actor: hospital agent. Trigger: a CFP arrives. Main flow: (1) the resource agent for the requested pool assesses the anonymised request against the hospital's own state; (2) the bidding policy decides what to report; (3) the agent replies PROPOSE with time to care, capability match and strain, or REFUSE with a reason code, within the bid window. Post-condition: the truthful assessment is retained privately for any later challenge; nothing about occupancy leaves the hospital.

UC5: Award and reserve. Actor: transfer manager. Precondition: deliberation is complete. Main flow: (1) bids that cannot meet the deadline are discarded; (2) the rest are scored by Eq. (15); (3) the winner's unit is reserved for 45 min; (4) the agreement and its terms hash are appended to the ledger; (5) the other bidders receive REJECT-PROPOSAL. Alternative flow: if the reservation race is lost to a concurrent award, the runner-up is tried.

UC6: Verify ledger and audit. Actor: reviewer or regulator. Main flow: (1) the reviewer requests verification; (2) the system recomputes every ledger block hash and Merkle root and every audit-row hash, and compares checkpoints; (3) it reports either an intact chain or the first broken row and the reason (link broken, contents changed, checkpoint mismatch).

UC7: Edit occupancy or policy. Actor: reviewer. Main flow: (1) the reviewer edits a pool's occupancy or a hospital's behaviour policy in the Hospital-data tab; (2) the API validates the value, and the database's `CHECK` constraint independently keeps occupancy within capacity; (3) the update trigger appends an audit row with before and after values and the actor; (4) the next negotiation reads the new state. Alternative flow: an attempt to change capacity or reference data is rejected by trigger, even for the database owner.

C. Class Model

Figure 9 shows the principal classes of the implementation, with their key attributes and operations.

D. Sequence Diagrams

Figure 10 traces one negotiation in which a protective hospital's false capacity refusal is challenged and struck out. Figure 11 traces the same transfer requested from the console and persisted to PostgreSQL.

E. User Interface

Figure 12 shows the live console. The *Live negotiation* tab runs transfers on the real Houston network, optionally with protective hospitals, and animates the six protocol steps as they stream in. The *Hospital data* tab edits occupancy and behaviour, each edit audited with before and after values and actor, and restores the reported data. The *Results* tab shows the generated dashboard. pgAdmin is bundled so that every table can be inspected and tampering can be attempted directly.

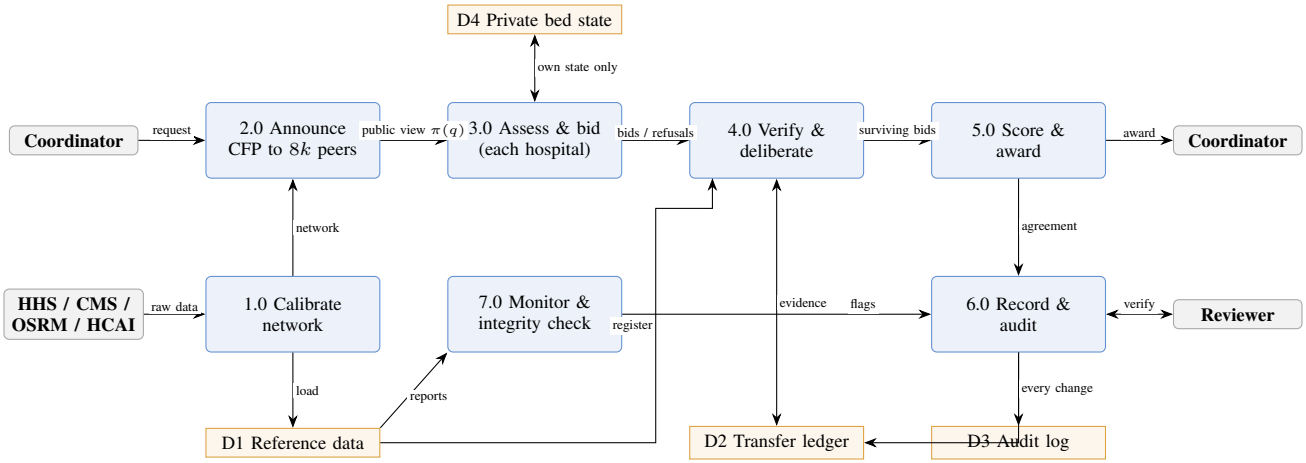


Fig. 6. Level 1 data flow diagram. D4 is held separately by every hospital agent and never leaves it; all other processes see only public data, bids and the ledger.

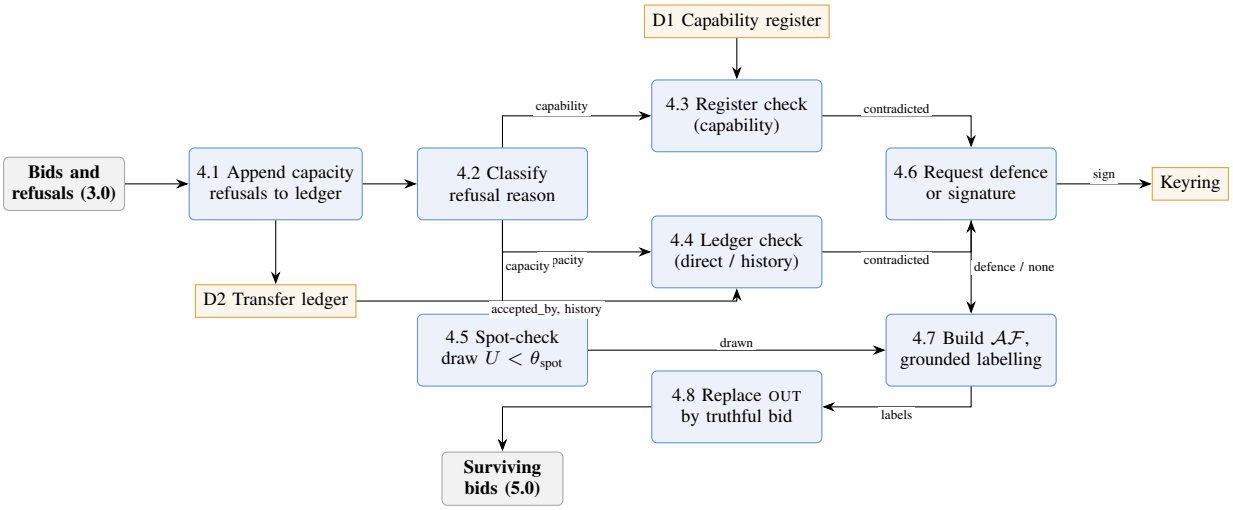


Fig. 7. Level 2 data flow diagram for process 4.0. Capability claims are checked against the register, capacity claims against the ledger; only contradicted or spot-checked refusals reach the defence step, and grounded labelling decides.

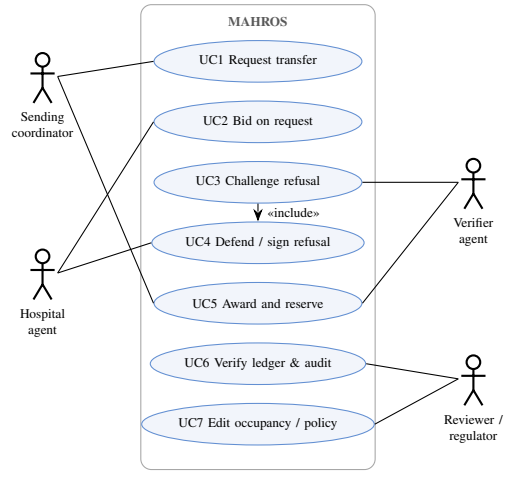


Fig. 8. Use-case diagram of MAHROS.

VII. Implementation

A. Software Organisation

The engine is a Python package whose modules correspond one-to-one to the layers of Fig. 2:

- `hospital/`: resource agents, hospital agent and behaviour policies.
- `negotiation/`: Contract Net, challenge engine, argumentation, attestation, collusion, scoring and message bus.
- `sim/`: real-network builder, warm start, scenarios, the five baselines, event loop and metrics.
- `calibration/`: vendored HHS, CMS, OSRM and HCAI data with their loaders.
- `ledger/`, `integrity/`, `fairness/`, `privacy/` and `eval/`: the hash-chained ledger, the Data Integrity rules, burden metrics, pseudonymisation and leakage model, and statistics with a Hungarian solver.
- `db/`: PostgreSQL schema, store and live synchronisation.
- `server/`: FastAPI application and the browser console.

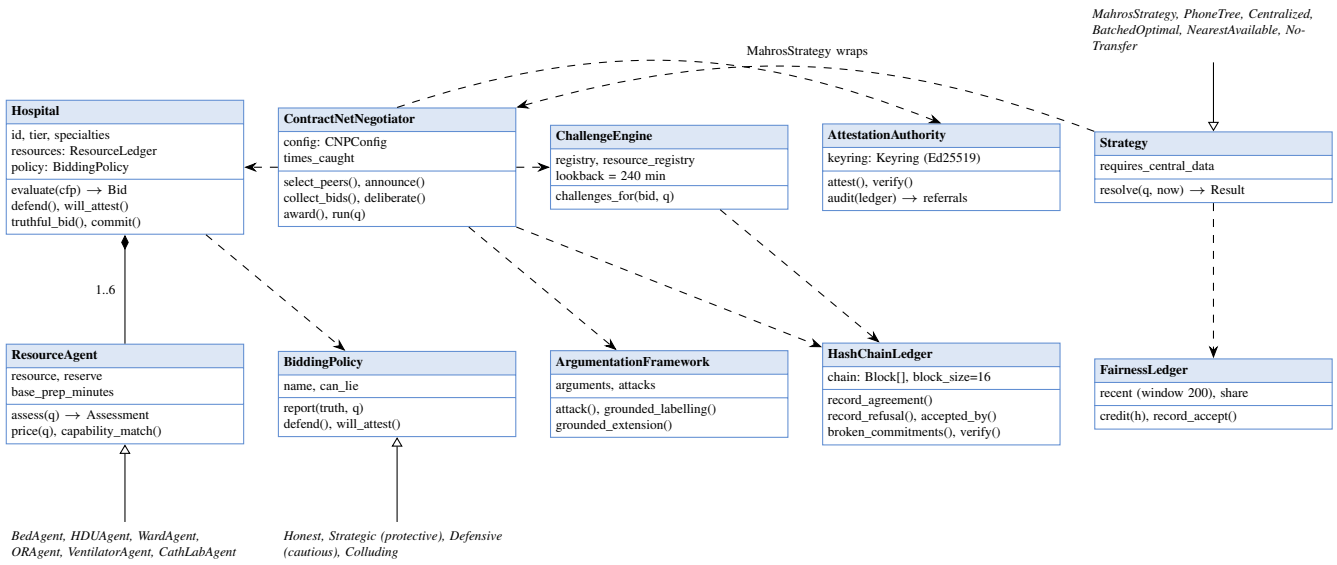


Fig. 9. Class diagram of the core engine (names as in the source code). Solid diamond: composition; dashed: uses; open triangle: specialisation.

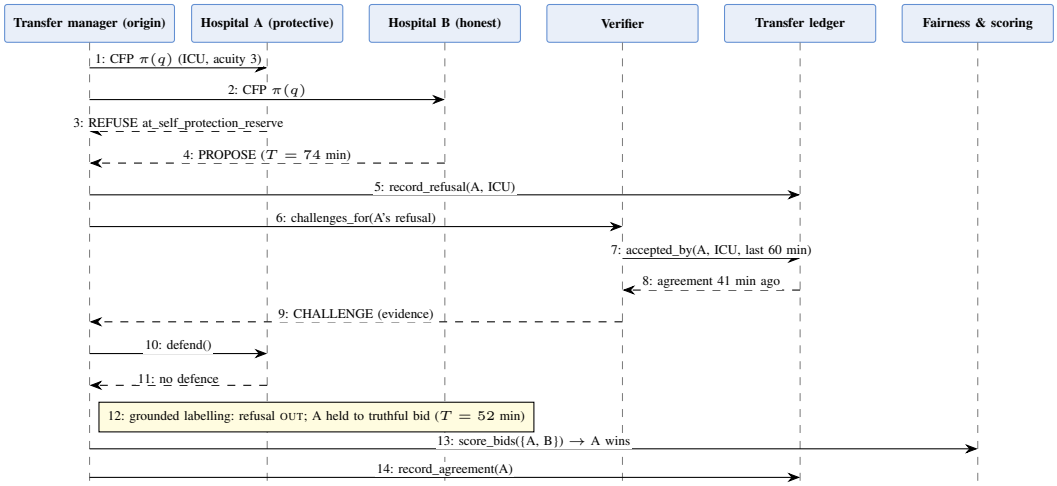


Fig. 10. Sequence of a negotiation with a challenged false refusal. Times are illustrative of the message order, not measured values.

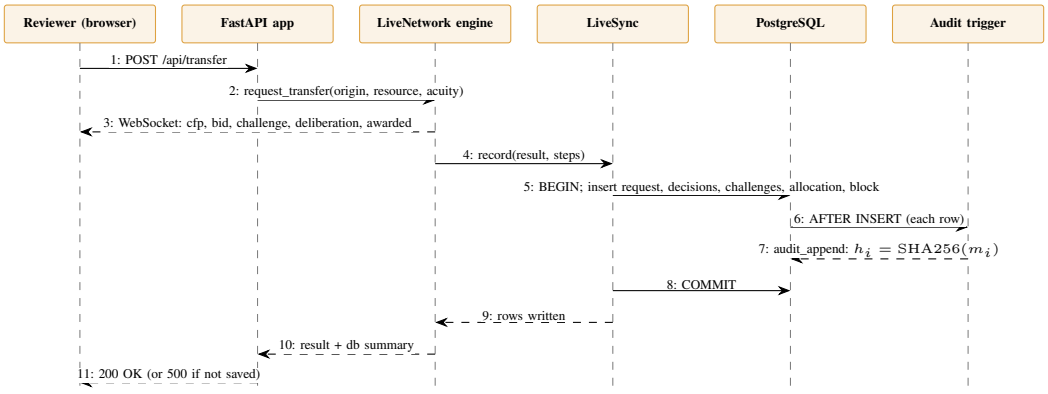


Fig. 11. Sequence of a console transfer persisted to PostgreSQL. The whole negotiation is one transaction; a failed write is reported rather than silently dropped.

The core uses only the Python standard library, so every experiment runs without building an environment. Each experiment is one script that writes a JSON result file. A build script regenerates the results document from those files, and the figures in this paper are drawn from data tables generated the same way, so no reported number is typed by hand.

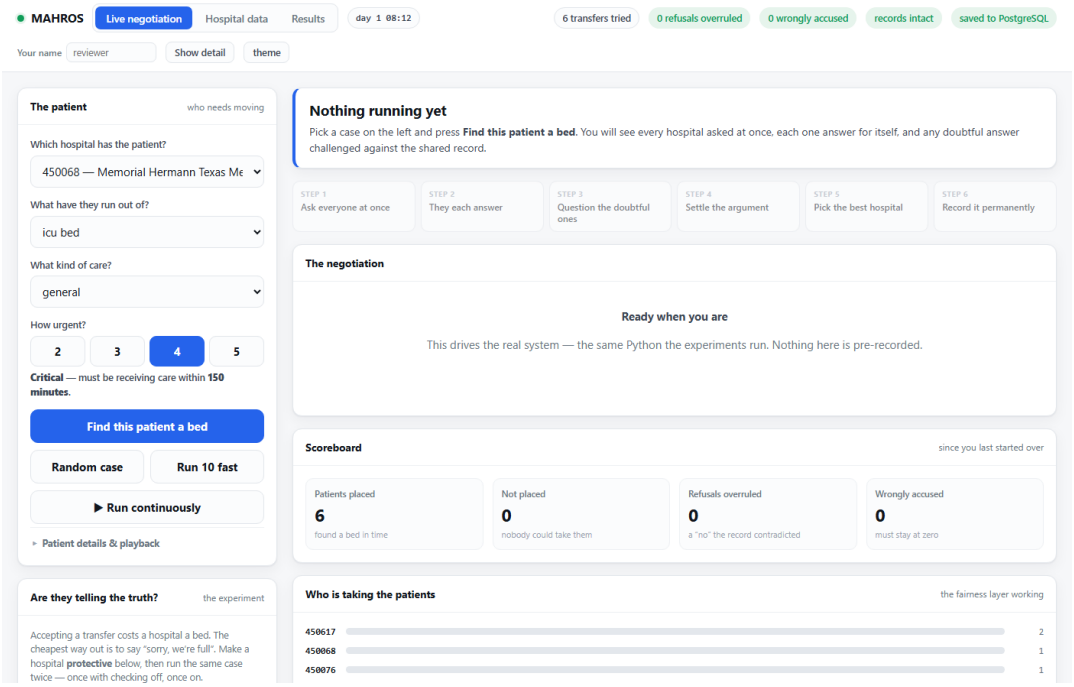


Fig. 12. The MAHROS live console (Live negotiation tab) connected to PostgreSQL. Left: the case to place. Centre: the six protocol steps and a scoreboard of placements, overruled refusals and wrongly accused hospitals. Top right: record integrity and persistence status.

B. Simulation Engine

The simulator is a discrete-event loop over a priority queue of timed events: admission, escalation, negotiation, care start, expiry, discharge and periodic sampling. Each run is fully determined by `(scenario, seed)`. All random streams, including hospital behaviour and spot-check selection, are seeded from SHA-256 digests rather than Python's per-process randomised `hash()`, and a regression test checks that a run reproduces exactly in a fresh process. Every strategy, MAHROS and baselines alike, implements one interface, `resolve(request, now)`, and runs in the same loop on the same patient stream, so differences in outcome come from the allocation logic alone.

C. Message Bus and Decentralisation Audit

All inter-agent traffic passes through a message bus that records FIPA-style performatives (CFP, PROPOSE, REFUSE, ACCEPT-PROPOSAL, REJECT-PROPOSAL) with sender, receiver and payload. Baselines log their traffic to the same bus, including every telephone answer, so all arms are audited on equal terms. The bus computes the decision concentration, the largest share of capacity-bearing messages received by any single node, and checks that no node obtains a global view.

Table X lists every message type and its exact payload. No field carries an occupancy count, a bed count or a patient identity.

TABLE X
PROTOCOL MESSAGES AND PAYLOADS

Performative	Direction	Payload fields
CFP	origin → peer	request id, origin, pseudonymous patient token, resource, specialty, acuity, created / expires time, expected stay
PROPOSE	peer → origin	feasible, time to care, capability match
REFUSE	peer → origin	feasible = false, reason code
ACCEPT-PROPOSAL	origin → winner	time to care
REJECT-PROPOSAL	origin → others	none
challenge / defence	within $\mathcal{AF}$	claim text, reason code, evidence (ledger ids)
attestation	peer → verifier	hospital, request, resource, predicate, time, nonce, two Ed25519 signatures

D. Privacy Measure

For a pool of capacity c, an observer's prior over occupancy is uniform on $\{0, \ldots, c\}$. A message consistent with a subset $\mathcal{S}$ of occupancy levels discloses

$$I = \log_2(c+1) - \log_2 |\mathcal{S}| \text{ bits.} \tag{18}$$

A MAHROS offer rules out only the levels at or above the reserve, about 0.06 bits on a 26-bed ICU. A capacity refusal bounds occupancy to the top few levels, about 3.2 bits. A centralised message carrying strain reveals all $\log_2(c+1)$ bits. Attestation adds nothing, because the signed predicate is exactly what the refusal already asserted. Patients are referenced only by an HMAC-SHA256 pseudonym, and free-text notes pass through a pattern-based scrubber for names, record numbers, phone numbers, dates and e-mail addresses before anything leaves the origin.

E. Data Integrity Rules

The Data Integrity agent applies hand-checkable rules to every real weekly report. A report is *red* (impossible as

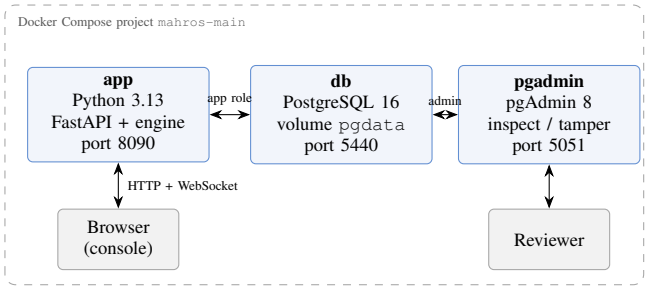


Fig. 13. Deployment: three containers started by one `docker compose up`. Data persist in a named volume; `down -v` reloads the real data.

reported) if any of the following holds:

- it shows more inpatients than staffed beds, more ICU patients than ICU beds, or more ICU beds than inpatient beds (with a 0.5 tolerance for 7-day averaging);
- it reports zero inpatients in a staffed hospital that reports patients in other weeks;
- its bed count is below half the median of the surrounding ± 8 weeks.

A report is *amber* (looks copied or stale) if the same non-zero inpatient count appears four or more weeks running, or if occupancy jumps by at least 30 points for one week and returns to within 10 points. Flags are stored in the `report_flags` table and shown per hospital in the console.

F. Deployment Architecture

Figure 13 shows the Docker Compose deployment. On start, the application container initialises the schema idempotently, loads the vendored real data through the privileged loader role, and serves the console and API through an unprivileged application role.

G. API Specification

Table XI lists the principal HTTP endpoints of the FastAPI application. A WebSocket at `/ws` streams the protocol steps of each live transfer.

H. Security, Privacy and Error Handling

The application role can write only the live columns it needs (`occupied`, `policy`, discharge fields) and insert records; it cannot modify reference data, records or the audit log. The audit and checkpoint functions run as security definers under an advisory lock, so concurrent writers cannot fork the chain. Each live negotiation, including its request, agent decisions, challenges, allocation and ledger block, is written in a single transaction, and a failed write is reported to the caller (HTTP 500) rather than silently dropped. If the database is unreachable, the console continues in memory and displays "not saving". Occupancy edits beyond capacity are rejected by the API and, independently, by a `CHECK` constraint. A transfer that loses a reservation race is re-awarded to the next bid, and a reservation that lapses before arrival triggers renegotiation. The optional language-model coordinator only writes explanations, is off by default, and never decides an award.

TABLE XI
PRINCIPAL API ENDPOINTS

Method	Endpoint	Purpose
POST	`/api/transfer`	run one negotiation; persist it
GET	`/api/state`, `/api/scenarios`	network state; available weeks
POST	`/api/occupancy`, `/api/policy`	edit live occupancy / behaviour
POST	`/api/fairness`, `/api/argumentation`	ablation switches
GET	`/api/hospitals`, `.../{id}/history`	hospital data and audited edits
POST	`/api/restore`, `.../{id}/restore`	restore reported data
GET	`/api/integrity`, `/api/data-integrity`	accountability; report flags
GET	`/api/bottlenecks`	Regional Monitor
GET	`/api/ledger`, `/api/ledger/verify`	ledger state; verification
GET	`/api/audit`, `/api/audit/verify`	audit log; chain verification
POST	`/api/audit/checkpoint`	anchor the chain head
POST	`/api/tamper`	demonstration: alter a ledger record
GET	`/api/transfers`, `/api/transfers/{id}`	transfer history; full record

TABLE XII
TAMPERING ATTEMPTS, CONTROLS AND VERIFYING TESTS

Attempt	Control	Verified by
App changes a capacity	column grant + trigger	DB test, app role: update rejected
Owner edits real data	`BEFORE` trigger	DB test, owner role: update rejected
App edits a record	insert-only grants	DB test: update/delete denied
Occupancy $>$ capacity	`CHECK` constraint	DB test with validation bypassed
Edit an audit row	hash recomputation	DB test: verify reports "contents changed"
Delete an audit row	chain link check	DB test: verify reports "link broken"
Truncate the log tail	checkpoint	DB test: checkpoint mismatch
Alter a signed refusal	Ed25519 verify	attestation test: every field
Forge the co-signature	independent key	attestation test: forgery rejected
Rewrite the whole chain	external checkpoint	procedural (outside the database)

Table XII maps each tampering attempt to the control that stops or detects it, and to the test that verifies the control.

I. Testing

Table XIII summarises the 205 test functions by area. Database tests run against a live PostgreSQL instance as both owner and application role, and an API smoke test exercises every database endpoint of a running deployment.

VIII. EXPERIMENTAL EVALUATION

A. Experimental Setup

Scenarios. Each run simulates 14 days of a real network in a real week (Table XIV). The Houston surge week (25 July 2021, Delta-wave peak, all 27 facilities reporting) is the primary scenario for the adversarial, attestation, collusion and sensitivity experiments.

TABLE XIII
TEST SUITE COVERAGE (205 TEST FUNCTIONS)

File	Tests	Representative properties
`test_argumentation`	29	reinstatement; even cycle undecided; honest never overruled; never compelled beyond capacity
`test_real_network`	38	real network build, capability join, Erlang fit, provenance
`test_core`	25	same seed same result in a fresh process; t-test vs. `scipy`; TOST; Hungarian
`test_db`	24	owner cannot change real data; edited or deleted audit row breaks chain; checkpoint catches truncation
`test_server`	23	API and console behaviour
`test_attestation`	21	any field tamper fails verification; co-signature unforgeable
`test_collusion`	18	abstention beats ledger; spot checks never punish honest refusals
`test_integrity_rules`	10	each red/amber rule on hand-built series
`test_ledger_index`	6	indexed lookups equal full scans
`test_case_mix` + `warm_start`	11	case-mix shares; length-biased remaining stay

TABLE XIV
SCENARIO WEEKS, SELECTED BY OBSERVED NETWORK OCCUPANCY

Scenario	Week	Selection rule
`houston_calm`	2020-08-23	calmest fully reported week
`houston_baseline`	2021-01-10	reference week
`houston_typical`	2022-05-08	median week of the series
`houston_surge`	2021-07-25	Delta-wave peak
`houston_post_pandemic`	2023-12-10	post-pandemic capacity crisis
`phoenix_calm/baseline`	2021-03-14 / 01-10	same rules, Phoenix,
`phoenix_typical/surge`	2022-07-17 / 2024-01-14	no parameter re-tuned

Baselines. All strategies run in the same event loop on the same network and the same patient streams. (i) *Phone tree*, today's practice: serial calls to the nearest peers, $\mathcal{N}(9,3)$ min per call, up to eight calls, one coordinator per 150 beds, and *perfect information* (no stale answers) in the headline comparison. (ii) *Central greedy*: reads every hospital's true private state and awards first-come-first-served after 1 min. (iii) *Batched optimal*: collects requests over 5 min and solves a minimum-cost assignment with the Hungarian method [15], again with full true state. (iv) *Shared registry*: picks the nearest hospital whose *declared* state shows room. (v) *No transfer*: the patient waits for a local unit.

Metrics. Success is the fraction of requests whose care starts within the clinical window (objective (1)). We also report mean and 90th-percentile time to care, critical-acuity miss rate, coordinator minutes per transfer, burden Gini over capacity-normalised acceptances, messages per transfer, decision concentration (the largest share of capacity-bearing messages any one node receives), and the adversarial counters: fabricated refusals, challenges, refusals overruled and *false accusations* (overrules of hospitals that never misreport).

Statistical protocol. Seeds are $42+101i$, and arms are paired by seed (common random numbers), so seed-to-seed variance cancels. For n paired differences Δ_i with mean $\bar{\Delta}$ and standard deviation s_Δ, the test statistic and effect size are

$$t = \frac{\bar{\Delta}}{s_\Delta/\sqrt{n}}, \qquad d_z = \frac{\bar{\Delta}}{s_\Delta}, \qquad \mathrm{CI}_{95} = \bar{\Delta} \pm t_{0.975,n-1}\frac{s_\Delta}{\sqrt{n}}. \tag{19}$$

The four primary comparisons on the surge week use 12 seeds and are corrected by Holm's method [40], which compares the j-th smallest of m p-values with $\alpha/(m-j+1)$. Equivalence with the optimisers is tested by the two one-sided tests procedure [41], [42] with a margin $\varepsilon = 2$ points declared before the analysis. Equivalence holds when both

$$t_{\mathrm{L}} = \frac{\bar{\Delta}+\varepsilon}{s_\Delta/\sqrt{n}} > t_{0.95,n-1}, \qquad t_{\mathrm{U}} = \frac{\bar{\Delta}-\varepsilon}{s_\Delta/\sqrt{n}} < -t_{0.95,n-1}, \tag{20}$$

which is equivalent to the 90% interval lying inside $(-\varepsilon,\varepsilon)$. Secondary experiments use 5–6 paired seeds. All statistics are implemented in pure Python, and the t distribution is cross-checked against `scipy` in the test suite.

Phone-tree model. For the telephone baseline, the j-th call ends at $t_j = t + \sum_{i\le j} D_i$ with $D_i \sim \max(2, \mathcal{N}(9,3))$ min. Each callee is assessed at its own call time t_j, so the hospitals reached late are assessed on a network that has moved on. The search stops at the first acceptance or after eight calls. It also requires a free coordinator at the origin, one per 150 beds.

B. Worked Example: One Real Negotiation Trace

Table XV reproduces one negotiation exactly as the live engine recorded it. The network is the Houston reference week. The six tertiary centres were set to PROTECTIVE, and this was the 83rd request of the session: an emergent (acuity 3, 240-minute window) general ICU request from St Luke's Hospital at the Vintage (CCN 670075), with an expected stay of 53.9 h. The CFP reached the eight nearest peers.

- Seven peers bid. Memorial Hermann Memorial City (450610), a protective tertiary centre, refused with `at_self_protection_reserve`.
- The Verifier found that 450610 had accepted another ICU transfer one minute earlier and raised a challenge citing that ledger entry. The hospital's defence failed, because its own state showed a free ICU bed. Grounded labelling struck the refusal out (16 arguments, 8 attacks, 8 IN, 8 OUT), and 450610 was held to its truthful bid of 64.3 min.
- Each of the seven bidders had recently taken more than its capacity share, partly because the protective centres had been refusing, so each raised a burden objection. Each objection was admissible because comparable alternatives existed and the patient was not critical, and each bidder withdrew.
- The award went to 450610 with score 0.882: time term 0.892, capability 1.0, strain term 0.975, and a fairness bonus because its recent share was 56% below its capacity share. Care was due in 64 min, leaving 175 min of slack.

The example shows both mechanisms acting together. Verification removed a false refusal, and the fairness layer sent the patient to the hospital that had been avoiding its share.

TABLE XV
ONE RECORDED NEGOTIATION (HOUSTON REFERENCE WEEK, REQUEST 83, ICU, ACUITY 3)

Peer (CCN)	Policy	Bid	T (min)	Outcome
450844	honest	propose	45.0	burden objection, withdrew
450638	honest	propose	57.3	burden objection, withdrew
450803	honest	propose	62.0	burden objection, withdrew
450610	protective	refuse	–	overruled; **awarded** (64.3)
450684	honest	propose	70.3	burden objection, withdrew
450644	honest	propose	70.5	burden objection, withdrew
670077	honest	propose	73.2	burden objection, withdrew
450035	honest	propose	74.0	burden objection, withdrew

TABLE XVI
COLD-START OCCUPANCY VALIDATION (5 SEEDS, 35% BURN-IN)

Scenario	n	MAE, reported	R^2	MAE, target	> 0.95
houston_baseline	27	3.83	0.954	2.17	15
houston_surge	27	5.64	0.813	3.54	19
phoenix_baseline	30	3.03	0.966	2.36	4

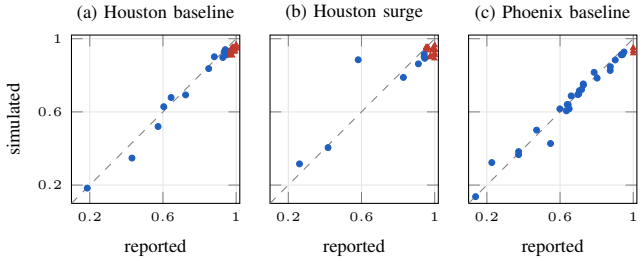


Fig. 14. Per-facility occupancy, simulated from an empty start vs. reported to HHS. Red triangles: facilities above the 0.95 ceiling.

C. Simulator Validation Against Real Occupancy (O6)

Validation starts from an *empty* network, so it tests where the fitted arrival rates lead, not where the run was placed (Table XVI, Fig. 14). In the Houston reference week the per-facility mean absolute error against reported occupancy is 3.83 points ($R^2 = 0.954$). Phoenix, fitted by the same code with nothing tuned, gives 3.03 points ($R^2 = 0.966$). In the Houston surge week the error is 5.64 points ($R^2 = 0.813$), concentrated in the 19 facilities that reported above the 0.95 fitting ceiling. The simulated surge network is therefore slightly less congested than the real one, which favours the baselines.

D. Reliability of the Real Capacity Reports

The Data Integrity agent examined every weekly report from both networks (Table XVII). In Houston, 306 of 5,673 reports (5.4%) are flagged, 223 of them as impossible, and 20 of 27 hospitals have at least one impossible report. The most common patterns are zero inpatients in a staffed hospital (149) and a bed count collapsing below half of the surrounding weeks (71). Phoenix's reports are far cleaner, with 0.2% flagged. Self-reported capacity from these same hospitals therefore cannot be assumed correct, which is the premise for verification.

E. Coordination on an Honest Network (O1, O2, O4)

Across all five Houston weeks, MAHROS places 4.7 to 13.0 points more patients in time than the phone tree, and the gap

TABLE XVII
DATA INTEGRITY AGENT APPLIED TO THE REAL HHS WEEKLY REPORTS

	Houston	Phoenix
Reports checked	5,673	6,169
Flagged (any)	306 (5.4%)	10 (0.2%)
Impossible as reported (red)	223 (3.9%)	5 (0.1%)
Hospitals with a red report	20 of 27	3 of 30
zero patients in staffed hospital	149	0
bed-count collapse	71	1
frozen count, 4+ weeks (amber)	66	3
one-week spike (amber)	17	2
patients > beds; ICU anomalies	3	4

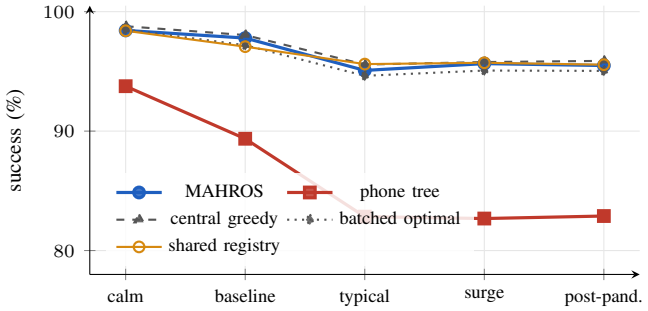


Fig. 15. Transfer success on the honest Houston network across five real weeks (5 seeds each).

TABLE XVIII
HONEST HOUSTON NETWORK: SUCCESS RATE (%), 5 SEEDS PER CELL

Week	MAHROS	Phone	Central	Batched	Registry	None
calm	98.4	93.8	98.8	98.5	98.4	42.4
baseline	97.8	89.4	98.0	97.2	97.1	45.7
typical	95.1	82.8	95.6	94.6	95.6	44.2
surge	95.7	82.7	95.8	95.1	95.7	47.8
post-pandemic	95.5	82.9	95.9	95.0	95.6	50.5

TABLE XIX
PRE-DECLARED PAIRED COMPARISONS, SURGE WEEK, 12 SEEDS, HOLM-ADJUSTED

MAHROS minus	Diff. (pts)	95% CI	Holm p	Equivalent (± 2)?
Phone tree	$+13.0$	$[+11.9, +14.1]$	< 0.0001	–
Central greedy	-0.3	$[-1.0, +0.4]$	0.744	yes ($p = 0.0001$)
Batched optimal	$+0.4$	$[-0.2, +1.1]$	0.497	yes ($p = 0.0001$)
Shared registry	-0.0	$[-0.6, +0.6]$	0.898	–

widens as the network fills (Fig. 15, Table XVIII). MAHROS, the two omniscient optimisers and the registry lie within about one point of each other every week. Without transfers only 42–51% of requests reach care in time.

Paired tests. On the surge week with 12 seeds (Table XIX), MAHROS exceeds the phone tree by 13.0 points (95% CI $[11.9, 14.1]$, $d_z = 7.4$, Holm $p < 10^{-4}$). It is equivalent within ± 2 points to central greedy (-0.3 points) and to batched optimal ($+0.4$ points), with TOST $p \approx 10^{-4}$ for both. The shared registry ties MAHROS on an honest network (-0.0 points, $p = 0.898$).

Mechanism and cost. The gain comes from parallelism. Eight hospitals answer within one 2-minute window, while the phone tree has spent about 36 minutes by its fifth call. The benefit concentrates in the sickest patients: in one surge-

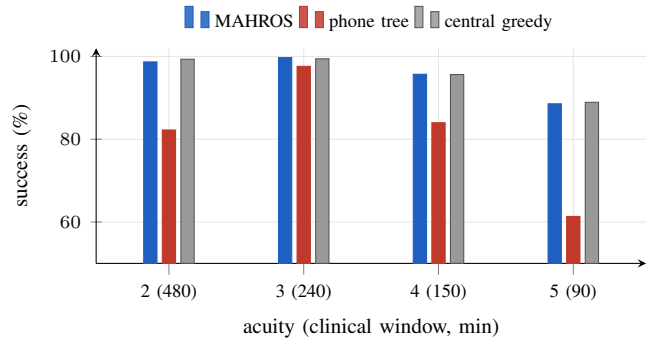


Fig. 16. Success by acuity, surge week, one seed (42), honest network. The shortest windows punish the serial search most; MAHROS matches the omniscient central arm.

TABLE XX
SURGE WEEK, FURTHER MEASURES (5 SEEDS)

	MAHROS	Phone	Central	Batched	Registry
Mean time to care (min)	65.3	87.2	60.4	65.1	64.0
90th pct. time to care (min)	83.1	133.3	79.1	82.9	81.4
Critical miss rate	7.3%	25.2%	7.0%	8.6%	7.3%
Coordinator min / transfer	1.1	50.3	0.0	0.0	0.0
Burden Gini	0.274	0.317	0.277	0.301	0.302
Messages / transfer	47.1	11.1	33.1	33.9	13.3
Decision concentration	0.11	0.11	1.00	1.00	1.00

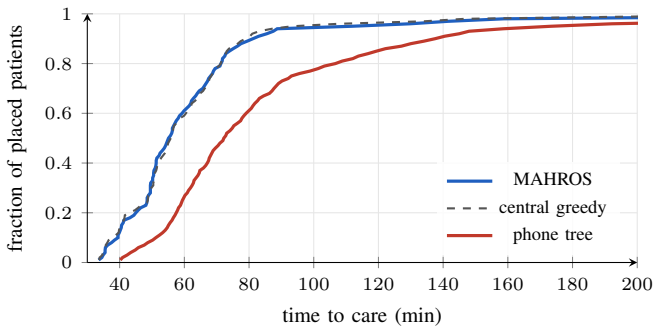


Fig. 17. Cumulative distribution of time to care for placed patients, surge week, one seed (42).

week seed, 89% of acuity-5 requests (90-minute window) were placed by MAHROS against 61% by the phone tree. Coordinator time falls from 50.3 to 1.1 min per transfer (Table XX). Decision concentration is 0.11, the same as the decentralised phone tree, against 1.00 for every central arm, and no message carries an occupancy value. The price of parallelism is message volume: 47 messages per transfer against 11.

Figure 17 shows the full distribution of time to care for placed patients. MAHROS and central greedy have nearly identical distributions (median 54.8 and 55.4 min), while the phone tree's median is 71.9 min and its 90th percentile 134 min.

Ablation and fairness. Removing the fairness term raises the burden Gini from 0.274 to 0.310 (Table XXI). The paired 12-seed test gives a Gini reduction of 0.034 ($p = 0.003$; bootstrap 95% CI $[0.017, 0.051]$) with no change in success (+0.1 points, $p = 0.79$). Figure 18 shows the effect per hospital:

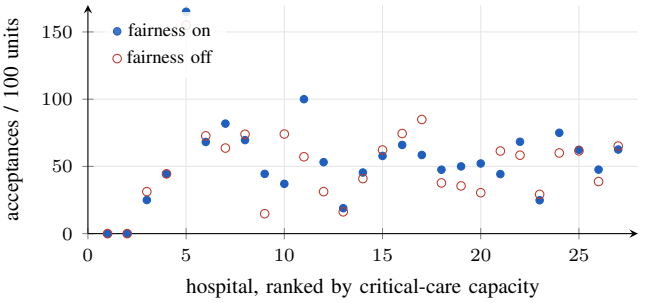


Fig. 18. Transfers accepted per 100 critical-care units (ICU, ventilator, cath lab, OR) for each of the 27 Houston hospitals, surge week, one seed (42), with and without the fairness layer.

TABLE XXI
ABLATION ON THE SURGE WEEK (5 SEEDS); ADVERSARIAL ARMS HAVE 50% PROTECTIVE HOSPITALS

Arm	Success	Burden Gini	Time to care
full	95.7%	0.274	65.3 min
no fairness	95.5%	0.310	62.3 min
no argumentation	96.3%	0.300	61.9 min
adversarial, full	86.2%	0.462	82.1 min
adversarial, no argumentation	64.5%	0.585	82.5 min
adversarial, no ledger	62.4%	0.586	85.4 min

TABLE XXII
SAFETY CHECK: NETWORKS WITH CAUTIOUS (TRUTHFUL, HIGH-RESERVE) HOSPITALS, COUNTS PER RUN

Network	Challenges	Overruled	False acc.
all honest	4,272	0.0	0
50% cautious	9,915	0.0	0
25% protective + 25% cautious	9,888	413.2	0

with the fairness layer, acceptances per unit of critical-care capacity are drawn towards a common level, especially among mid-sized hospitals. On an honest network, deliberation costs a non-significant 0.7 points of success and 2.0 minutes of time to care. Its value appears under misreporting.

F. Verification Under False Refusals (O3)

As the share of protective hospitals rises, success without checking falls from 96.4% to 26.6%. With MAHROS' verification it falls only to 80.5%, recovering 62–77% of the loss (Fig. 19, Table XXIII). At 50% protective, checking raises success from 63.7% to 86.5%. Removing the ledger costs as much as removing argumentation (62.4% against 64.5%, Table XXI), because the ledger is the evidence base for capacity claims. **No experiment produced a false accusation.** This held even on a network in which half the hospitals are cautious and attract more than twice the challenges (9,915 against 4,272 per run): every one of their refusals was defended, because each was true (Table XXII). Verification catches individual lies selectively: at 50% protective, 520.8 of 7,898 fabricated refusals per run are overruled. Yet 70% of the lost placements are recovered, because a request fails only when every asked hospital refuses, so overruling one false refusal per failing request is usually enough.

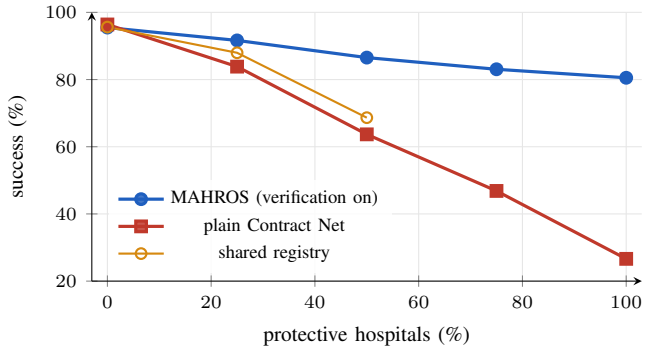


Fig. 19. Success vs. share of protective hospitals, surge week, 6 seeds (registry run at 0–50%).

TABLE XXIII
DOSE-RESPONSE, SURGE WEEK, 6 SEEDS (COUNTS PER RUN)

Protective	Off	On	Recovered	False ref.	Overruled	False acc.
0%	96.4%	95.5%	–	0	0.0	0
25%	83.8%	91.6%	62%	3,657	342.0	0
50%	63.7%	86.5%	70%	7,898	520.8	0
75%	46.8%	83.1%	73%	10,700	648.2	0
100%	26.6%	80.5%	77%	14,864	784.5	0

TABLE XXIV
SHARED REGISTRY VS. MAHROS UNDER IDENTICAL PROTECTIVE HOSPITALS (6 PAIRED SEEDS)

Network	Protective	Registry	MAHROS	Difference
Houston surge	0%	95.6%	95.5%	-0.1 ($p = 0.757$)
	25%	88.0%	91.6%	$+3.7$ ($p = 0.016$)
	50%	68.6%	86.5%	$+17.9$ ($p = 0.0002$)
Phoenix surge	0%	96.8%	96.9%	$+0.1$ ($p = 0.839$)
	25%	93.2%	94.9%	$+1.8$ ($p = 0.099$)
	50%	83.5%	92.2%	$+8.7$ ($p = 0.0016$)

Against a shared registry. With identical protective hospitals, the registry falls to 68.6% at 50% protective in Houston while MAHROS holds 86.5% (+17.9 points, $p = 0.0002$). In Phoenix the corresponding figures are 83.5% and 92.2% (+8.7 points, $p = 0.0016$; Table XXIV). A registry removes a hospital that declares itself full and never tests the claim.

G. Signed Refusals

With signed refusals at 50% protective hospitals, success rises from 62.3% when no liar is deterred ($\delta = 0$) to 82.0% at $\delta = 0.25$ and 85.6–86.7% for $\delta \geq 0.5$ (Fig. 20). Full deterrence reproduces the unattested protocol exactly. False accusations remained zero throughout, and the post-hoc audit-referral false-positive rate was 1.10%. Because deterrence is an institutional quantity that no dataset measures, the whole range is the result. Even modest deterrence recovers most of the loss (Table XXV). As δ rises, fewer false refusals are signed and more are withdrawn, so the number of signatures falls from about 21,000 to 9,900 per run.

H. Collusion and Spot Checks

A cartel that abstains in concert leaves no contradiction and cuts success from 95.5% to 89.1%, 79.2% and 70.5% for 3, 6 and 9 members (Fig. 21(a)). Co-refusal lift did not separate the cartel from an honest network under surge (maximum follow-on lift 1.10 against 1.24), because genuine scarcity makes honest hospitals refuse together. Uniform random spot checks did separate them. Checking 15% of capacity refusals restored success under a six-member cartel to 90.8% (+11.6 points, 95% CI $[8.1, 15.0]$, $p = 0.0004$), at no measurable cost on an honest network and with zero false accusations (Fig. 21(b)). Escalating checks on hospitals already caught reached 91.3%, which is not significantly better than uniform 15% ($p = 0.66$) and used 111% of the checks, so uniform random auditing is the recommended configuration (Table XXVI).

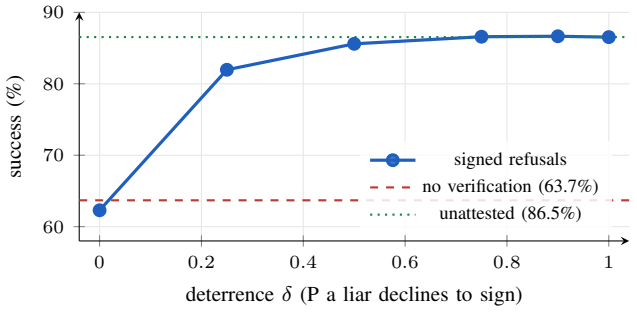


Fig. 20. Signed refusals at 50% protective hospitals, surge week, 6 seeds, Ed25519.

TABLE XXV
ATTESTATION SWEEP AT 50% PROTECTIVE (PER-RUN MEANS, 6 SEEDS)

Deterrence δ	Success	Signed	Declined	Audit referrals
0.00	62.3%	21,115	0	107
0.25	82.0%	13,889	370	186
0.50	85.6%	10,627	480	155
0.75	86.6%	9,966	538	121
0.90	86.7%	9,950	599	119
1.00	86.5%	9,900	622	108

TABLE XXVI
UNIFORM VS. RISK-BASED SPOT CHECKS (SIGNED PROTOCOL, 6 SEEDS)

Arm	Honest	6 colluding	Checks	Cartel overruled
no spot checks	95.5%	80.1%	0	65.7
uniform 5%	95.5%	88.6%	1,523	345.7
uniform 10%	95.5%	89.9%	2,571	414.8
uniform 15%	95.5%	90.8%	3,589	453.0
5%, 50% once caught	95.5%	91.3%	3,976	616.3
5%, 100% once caught	95.5%	91.3%	7,010	738.2

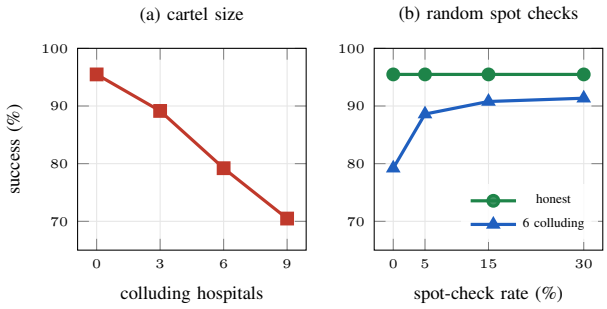


Fig. 21. Collusion under the signed protocol, surge week, 6 seeds.

TABLE XXVII
PHOENIX REPLICATION, 30 REAL HOSPITALS, NOTHING RE-TUNED (5 SEEDS)

Week	MAHROS	Phone	Registry	Central	Gain (pts)
calm	98.8%	88.6%	98.9%	98.8%	+10.2
baseline	98.7%	88.8%	98.3%	98.4%	+9.9
typical	98.6%	85.9%	98.2%	97.9%	+12.6
surge	96.9%	82.6%	96.7%	97.1%	+14.3

TABLE XXVIII
SENSITIVITY SWEEPS, SURGE WEEK, 5 SEEDS: SUCCESS (%) OF MAHROS / PHONE TREE AT EACH END OF THE RANGE

Parameter	Low	High	At low	At high
Clinical-window scale	0.70	1.50	86.3 / 66.8	97.5 / 90.6
Escalation probability	0.07	0.21	98.5 / 89.7	89.3 / 74.7
Handover allowance	8 min	20 min	97.0 / 83.6	92.6 / 79.5
Escalation delay	4 h	12 h	95.5 / 81.7	95.8 / 83.8
Cardiac cath share	0.25	0.75	93.7 / 80.4	96.6 / 84.2
Phone call duration	5 min	15 min	– / 86.5	– / 77.0
Phone staleness	0	0.40	– / 82.7	– / 73.5
Agent coordination cost	0.25	2 min	95.6 / –	95.6 / –

I. Replication on Phoenix

With no parameter re-tuned, MAHROS exceeds the phone tree by +10.2, +9.9, +12.6 and +14.3 points across the four Phoenix weeks (all $p < 10^{-4}$; Table XXVII) and again matches the registry and central greedy within about half a point. Under 50% protective hospitals in the Phoenix surge week, verification restores success from 79.7% to 92.5% (+12.8 points, $p = 0.0019$, 75% of the loss), and signed refusals give 91.6%. Phoenix ran cooler than Houston (4 facilities above the fitting ceiling against 15), so the replication tests a different congestion regime, and the effects are correspondingly smaller.

J. Sensitivity Analysis

Every unsourced parameter was swept on the surge week: clinical-window scale (0.7–1.5), call duration (5–15 min), phone staleness (0–0.40), agent coordination cost, deliberation time, adversary strength, escalation delay and probability, handover time and cardiac catheterisation share. MAHROS beat the phone tree at every value of every parameter (Fig. 22), and the gap widened as windows tightened (86.3% against 66.8% at 0.7 scale). Verification adds 21.8 points when protective hospitals misreport consistently ($\phi = 1$). When they misreport only half the time ($\phi = 0.5$), a second negotiation round already reaches them on a truthful turn, and verification changes success by at most −0.4 points (Table XXIX). The mechanism is therefore aimed at systematic misreporting, which is the case a registry cannot handle. The phone tree degrades monotonically with call duration (86.5% at 5 min, 77.0% at 15 min) and with stale information (82.7% at 0, 73.5% at 0.40), so the perfect-information headline is conservative. Agent coordination cost and deliberation time (0–5 min) changed nothing measurable. Table XXVIII gives the extremes of every sweep.

TABLE XXIX
ADVERSARY STRENGTH, 50% PROTECTIVE, SURGE WEEK, 5 SEEDS: SUCCESS WITH VERIFICATION OFF / ON (%)

Comfort threshold θ	0.30	0.55	0.80
shirk $\phi = 0.5$	95.7 / 95.7	95.9 / 95.5	95.9 / 95.8
shirk $\phi = 1.0$	64.5 / 86.2	64.5 / 86.2	64.5 / 86.2

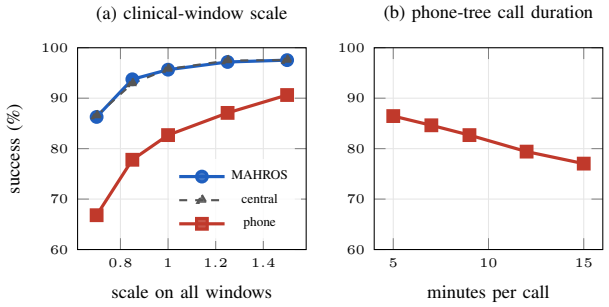


Fig. 22. Sensitivity on the surge week (5 seeds). The headline uses 9 min per call and zero staleness.

TABLE XXX
PER-TRANSFER DECISION TIME (MS), SURGE WEEK, 3 SEEDS

Configuration	n	p50	p95	p99
Plain Contract Net	1,080	0.62	2.04	3.31
+ ledger-backed verification	1,069	3.55	12.52	17.25
+ attestation ($\delta = 0.8$)	1,069	4.26	15.53	25.80
Adversarial, 50% protective	1,968	13.90	31.14	42.51

TABLE XXXI
OBJECTIVES (SECTION III-B) VS. MEASURED RESULTS

ID	Target	Measured	Met
O1	≥5 pts over phone tree	+13.0 pts, $p < 10^{-4}$	Yes
O2	equivalent within ±2 pts	TOST $p = 0.0001$ vs. both optimisers	Yes
O3	≥50% loss recovered, 0 false acc.	70% at 50% protective; 0 false acc.	Yes
O4	no occupancy sent; low concentration	none sent; concentration 0.11 vs. 1.00	Yes
O5	real data locked; edits detected	database tests pass (owner and app role)	Yes
O6	MAE < 5 pts, both cities	3.83 (Houston), 3.03 (Phoenix)	Yes
O7	p99 < 1 s	42.5 ms worst configuration	Yes

K. Decision Latency (O7)

Measured per transfer in a single process (Table XXX), the median decision time is 0.62 ms for plain Contract Net, 3.55 ms with ledger-backed verification, 4.26 ms with attestation and 13.9 ms in the adversarial setting. The 99th percentiles are 3.3, 17.3, 25.8 and 42.5 ms, all well inside the 1 s budget.

L. Objective Verification

Table XXXI summarises each objective against its measured result.

IX. DISCUSSION

A. Significance of the Results

The results support the central design claim: a hospital's own recorded commitments carry enough evidence to verify capacity refusals without anyone seeing capacity. Under

systematic false refusals, MAHROS recovered 62–77% of the lost placements and never overruled a truthful hospital, and it outperformed a shared registry by up to 17.9 points. On an honest network it matched two omniscient optimisers within the declared ± 2-point margin. The advantage over the telephone search comes from parallelism. What decentralised verification adds beyond any parallel method is robustness to misreporting while occupancy stays private.

B. Relationship to the Research Gaps

Gap 1 (the disclosure dilemma) is closed by combining parallel Contract-Net bidding with evidence-based checking, so that no occupancy value is ever transmitted. Gap 2 (no evidence base for private claims) is closed by the capability/capacity split and the ledger predicate of Eq. (10). Gap 3 (misreporting outside the evaluation) is addressed by the swept adversary, the cautious-hospital control and two real networks. Gap 4 (no inspectable accountability) is addressed by the Merkle-linked ledger, the trigger-enforced audit chain and the console, through which a reviewer can trace any award to its evidence.

C. Comparison with Alternatives

Compared with centralised optimisation [12], [13], [51], MAHROS gives up at most a fraction of a point of allocation quality while removing the need for a central occupancy store, and it models misreporting rather than assuming truth. The approaches are complementary: a regional optimiser could consume MAHROS' verified offers instead of raw self-reports. Compared with argumentation-based negotiation [28], [44], MAHROS admits only evidence-backed attacks, so outcomes never depend on persuasion and a truthful participant always prevails. Compared with reputation systems [31], [32], it needs no outcome feedback, which this domain never provides. Indeed, a minimal reputation mechanism (risk-based spot checks) did not outperform uniform random audits.

D. Trade-Offs

Parallel asking generates about four times the messages of a telephone search. Electronically this is negligible, but each message is a query to a hospital's bed system. Cautious hospitals face more challenges and always win them, which is friction rather than harm. The capacity predicate rests on a 60-minute window: a discharge inside the window can make a true refusal look contradicted, and the signed variant is the realistic way to let a hospital certify its position. Fairness reduced burden inequality significantly at no cost in success, and its guardrails prevent it from ever delaying a critically ill patient.

E. Deployment Implications

A regional deployment needs five elements: an interface from each hospital's bed-management system to its agent, a ledger operated by a party no single hospital controls, per-institution signing keys with an independent co-signer such as a state health department, governance for overruled refusals and audit referrals, and clinical sign-off on windows, reserves and weights. Statewide transfer lines show that cross-system coordination can be stood up quickly [11]. MAHROS is best positioned as the verification layer such a service adopts.

F. Generalisability

The protocol requires only that capability be public and commitments be recorded; nothing in it is specific to U.S. hospitals or to ICU beds. Any setting where autonomous providers allocate scarce, privately observed capacity and have an incentive to refuse has the same structure, including long-term-care placement, organ-transport logistics, shelter beds and specialist referral queues. Phoenix reproduced the direction of every effect with smaller magnitudes, as expected for a less congested region.

G. Ethical Considerations

No real patient data are used. All patients are simulated and labelled as such by a database constraint, and the hospital data are public and aggregate. The Data Integrity flags concern real reports from named hospitals. They are described as *unreliable*, never as dishonest, because a frozen or impossible report can have an innocent cause such as a stale data feed. The simulated misreporting policies are assigned to the largest hospitals first because those have most to gain; this is a modelling choice about incentives, not an allegation about any institution. An overruled refusal or an audit referral is designed as a prompt for review, never as an automatic sanction.

H. Limitations and Threats to Validity

Simulated behaviour. No public dataset records how hospitals answer transfer requests, so refusal behaviour is modelled and every behavioural parameter is swept. The occupancy model is validated against real reports; the behaviour model is not. *Proxy outcome.* Reaching care within an acuity-dependent window is a proxy for harm, not a mortality model, and the windows are design assumptions whose sweep preserved every ranking. *Data.* The HHS facility-level reporting mandate ended in May 2024, and the data exclude Veterans Affairs and other federal facilities. Road times are free-flow. The case-mix is Californian because no Texas or Arizona transfer data by diagnosis are public. *Tuned design choices.* The scoring weights and protocol timings are design choices. The fairness ablation covers one term, but the weights are not swept jointly. *Deterrence and collusion.* Signed refusals depend on an unmeasured deterrence parameter. Collusion is countered by random audits, not by detection. *Integrity.* Trigger locks prevent accidental change, and the audit chain detects deliberate change. A superuser rewrite is detected only by a checkpoint held outside the database.

X. Conclusion and Future Work

This paper presented MAHROS, a Multi-Agent Hospital Resource Optimization System that brokers inter-hospital escalation transfers by parallel Contract-Net negotiation among hospital agents that keep their occupancy private. Every refusal

becomes a verifiable claim: capability against the public CMS register, capacity against the refuser's own ledger-recorded commitments. Disputes are settled by Dung's grounded semantics, signed refusals and random spot checks extend verification to stronger adversaries, and every decision is recorded in a Merkle-linked ledger and a trigger-enforced, hash-chained PostgreSQL audit log.

Evaluated on 27 real Houston and 30 real Phoenix hospitals, with a simulator that reproduces reported occupancy within 3.8 and 3.0 points from a cold start, MAHROS met all seven objectives. It placed 13.0 points more patients in time than a perfect-information telephone search and was equivalent within ± 2 points to optimisers with full knowledge of occupancy. It recovered 62–77% of the placements lost to false refusals with zero false accusations, outperformed a shared registry by up to 17.9 points under misreporting, restored most of the loss caused by a six-hospital cartel with 15% random spot checks, and decided each transfer within milliseconds.

Future work will proceed in five directions:

1) calibrating refusal behaviour to transfer-centre logs under a data-use agreement;
2) estimating the deterrence effect of signed refusals through a field or survey study with bed managers;
3) replacing the clinical-window proxy with a delay–outcome model estimated from linked transfer and outcome data;
4) extending the privacy model to observers that accumulate messages over time, and evaluating secure computation for the award step [36];
5) a pilot integration with a regional transfer service in which a central optimiser consumes MAHROS-verified offers.

APPENDIX A
MODEL AND PROTOCOL PARAMETERS

Table XXXII lists every protocol, behaviour and baseline parameter with its value and status. *Swept* parameters were varied in the sensitivity analysis of Section VIII-J.

TABLE XXXII
PROTOCOL, BEHAVIOUR AND BASELINE PARAMETERS

Parameter	Value	Status
Clinical windows, acuity 1–5	720/480/240/150/90 min	design; scale swept 0.7–1.5
Peers per round	8 (16 in round 2)	design
Rounds; retry interval	2; 20 min	design
Bid window; deliberation	2 min; 0.75 min	design; deliberation swept 0–5
Reservation hold	45 min	design
Self-protection reserve	ICU 1, ward 2	design; 0 for acuity ≥ 4
Score weights	0.45, 0.25, 0.15, 0.15	design
Fairness window; objection	200 accepts; 0.35	design
Comparable-time band	15 min	design
Reserve-challenge window	60 min	design
Ledger look-back	240 min	design
Ledger block size	16 entries	design
Agent coordination cost	0.5 min / round	swept 0.25–2
Escalation probability	0.14	cited; swept 0.07–0.21
Escalation delay	lognormal, mean 7 h	swept 4–12 h
Handover allowance	12 min	swept 8–20
Cardiac cath share	0.5	swept 0.25–0.75
Comfort threshold θ	0.55	swept 0.3–0.8
Shirk probability ϕ	1.0	swept 0.5, 1.0
Expensive stay	$\geq$24 h	design
Cautious extra reserve	2 units	design
Deterrence δ	0.8 (default)	swept 0–1
Audit-referral window	5 min	chosen by measurement
Cartel abstention window	60 min	matches challenge window
Spot-check rate	0 (default)	swept 0–30%
Phone call duration	$\mathcal{N}(9, 3)$, min 2	swept 5–15
Phone calls before giving up	8	design
Phone stale-information prob.	0 (headline)	swept 0–0.40
Central / registry dispatch	1 / 3 min	design
Batch window (optimal)	5 min	design
Fitting ceiling $\bar{\rho}$	0.95	design
Simulated horizon	14 days	design

APPENDIX B
ATTACK RULEBOOK AND VERDICT RULES

Let $subj(A)$ be the hospital that argument A concerns. The only attacks are:

1) $\textsc{Challenge}(C) \wedge \textsc{Refusal}(R) \wedge subj(C) = subj(R) \Rightarrow C \to R$;
2) $(\textsc{Defence}(D) \vee \textsc{Attestation}(D)) \wedge \textsc{Challenge}(C) \wedge subj(D) = subj(C) \Rightarrow D \to C$;
3) $\textsc{BurdenObjection}(O) \wedge \textsc{Bid}(B) \wedge subj(O) = subj(B) \Rightarrow O \to B$;
4) $\textsc{CounterOffer}(K) \wedge \textsc{Bid}(B) \wedge subj(K) = subj(B) \Rightarrow K \to B$.

Hospitals never attack each other directly, so no even attack cycle arises and every argument is labelled IN or OUT. The fixed point of Eqs. (11)–(12) is reached in at most three passes. A refusal is IN if it is unchallenged or every challenge is answered, and OUT otherwise. For a refusal by h labelled OUT: if h's truthful assessment is feasible and meets the deadline, the truthful bid replaces the refusal and h is counted as caught; otherwise the refusal stands with its truthful reason. A bid labelled OUT through rule 3 becomes a withdrawal (`withdrew_over_fair_share`), recorded separately from an inability. A spot check is drawn only for capacity refusals, only under attestation, with probability θ' for a hospital caught before (if $\theta' > \theta_{\text{spot}}$) and θ_{spot} otherwise. A hospital that declines to sign a spot-checked refusal receives an unanswered challenge.

APPENDIX C
DATABASE SCHEMA

Table XXXIII summarises the PostgreSQL schema by table class and the rule that governs each.

APPENDIX D
REPRODUCTION

The repository at https://github.com/ROHITH-V-S/MAHROS (commit `1db8df5`) contains all code and the vendored real data. Every result is regenerated by one script per experiment:

```
pip install -e ".[demo,db,analysis,attest,dev]"
cd experiments
python validate_occupancy.py houston_baseline 5
python validate_occupancy.py houston_surge 5
python validate_occupancy.py phoenix_baseline 5
python data_integrity.py
python run_all.py 5
python significance.py 12
```

TABLE XXXIII
POSTGRESQL TABLES BY CLASS

Class	Tables	Rule
Reference (real)	`hospitals`, `hospital_capabilities`, `hhs_weekly_reports`, `travel_times`	no app grant; `BEFORE` trigger blocks change except during data load
Live state	`resources` (`occupied` only), `hospital_settings` (`policy` only)	column grants; capacity locked by trigger; `CHECK` keeps occupancy within capacity; audited
Records	`live_epochs`, `patients`, `transfer_requests`, `agent_decisions`, `challenges`, `allocations`, `admissions`, `ledger_blocks`, `ledger_entries`, `report_flags`	insert-only for the app (admission discharge columns updatable); audited
Experiment	`simulation_runs`, `phone_tree_records`, `prediction_records`	run metadata and outputs
Integrity	`audit_log`, `audit_checkpoints`, `system_events`	appended only through security-definer functions

```
python adversarial.py 6
python registry_attack.py 6
python attestation.py 6
python collusion.py 6
python reputation.py 6
python replication.py 5
python sensitivity.py 5
python realtime.py
python build_results.py
cd .. && python -m pytest
```

The live system starts with `docker compose up -d --build`; the console is served on port 8090 and pgAdmin on port 5051. Seeds are $42 + 101i$, and every run is fully determined by `(scenario, seed)`.

ACKNOWLEDGMENT

[Acknowledgment placeholder.] Hospital data are U.S. public-domain reports (HHS, CMS); road times derive from OpenStreetMap (ODbL) via OSRM; the transfer case-mix is California HCAI open data (CC-BY).

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